

01_MME_Onset

April 30, 2025

#####



#

AGHYMET RCC-WAS

© Mandela HOUNGNIBO 2024 (Find more [here](#))

0.1 Library

Set forecast working directory

Set Climatological Years

0.2 Download Observation and Process

Load Downloader

Data Download Area



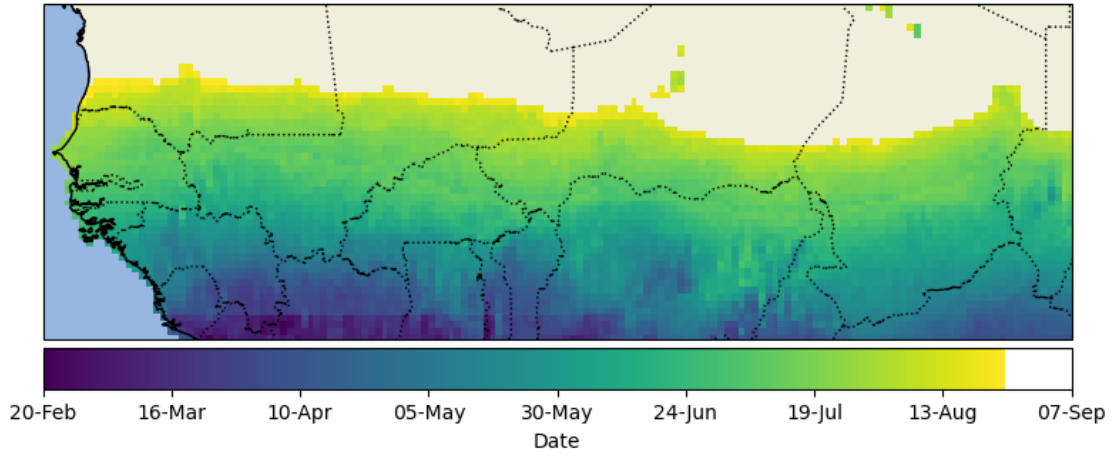
Agro_PRESASS_2025/Observation/Daily_PRCP_1991_2024.nc already exists. Skipping download.

Agro_PRESASS_2025/Observation/Daily_PRCP_2025_2025.nc already exists. Skipping download.

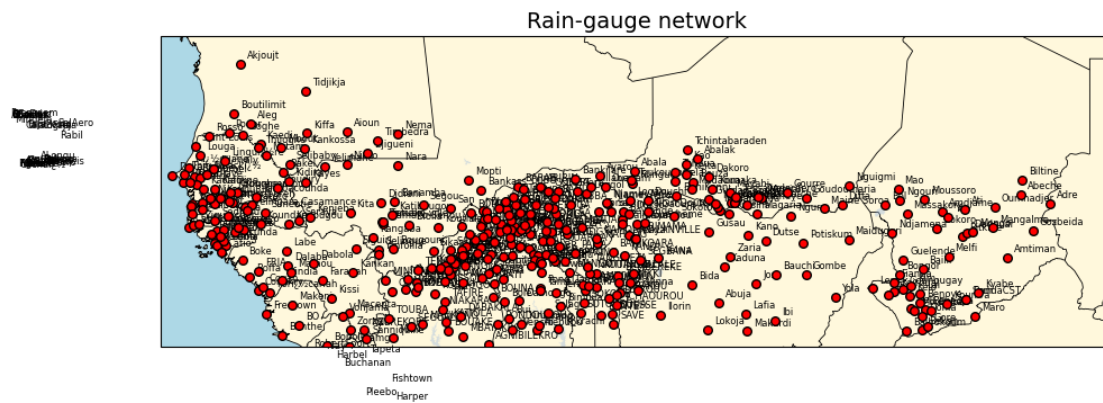
0.3 Compute Onset

```
{0: {'zone_name': 'Sahel100_0mm',
    'start_search': '06-01',
    'cumulative': 15,
    'number_dry_days': 25,
    'thrd_rain_day': 0.85,
    'end_search': '08-30'},
1: {'zone_name': 'Sahel200_100mm',
    'start_search': '05-15',
    'cumulative': 15,
    'number_dry_days': 25,
    'thrd_rain_day': 0.85,
    'end_search': '08-15'},
2: {'zone_name': 'Sahel400_200mm',
    'start_search': '05-01',
    'cumulative': 15,
    'number_dry_days': 20,
    'thrd_rain_day': 0.85,
    'end_search': '07-31'},
3: {'zone_name': 'Sahel600_400mm',
    'start_search': '03-15',
    'cumulative': 20,
    'number_dry_days': 20,
    'thrd_rain_day': 0.85,
    'end_search': '07-31'},
4: {'zone_name': 'Soudan',
    'start_search': '03-15',
    'cumulative': 20,
    'number_dry_days': 10,
    'thrd_rain_day': 0.85,
    'end_search': '07-31'},
5: {'zone_name': 'Golfe_Of_Guinea',
    'start_search': '02-01',
    'cumulative': 20,
    'number_dry_days': 10,
    'thrd_rain_day': 0.85,
    'end_search': '06-15'}}}
```

Compute onset for gridded products



Process Onset in combining with Ground-Observation



Year = 1991-02-01T00:00:00.000000000, number in-situ = 391

Year = 1992-02-01T00:00:00.000000000, number in-situ = 378

Year = 1993-02-01T00:00:00.000000000, number in-situ = 387

Year = 1994-02-01T00:00:00.000000000, number in-situ = 388

Year = 1995-02-01T00:00:00.000000000, number in-situ = 386

Year = 1996-02-01T00:00:00.000000000, number in-situ = 393

Year = 1997-02-01T00:00:00.000000000, number in-situ = 384

Year = 1998-02-01T00:00:00.000000000, number in-situ = 379

Year = 1999-02-01T00:00:00.000000000, number in-situ = 382

Year = 2000-02-01T00:00:00.000000000, number in-situ = 384

Year = 2001-02-01T00:00:00.000000000, number in-situ = 378

Year = 2002-02-01T00:00:00.000000000, number in-situ = 379

Year = 2003-02-01T00:00:00.000000000, number in-situ = 387

Year = 2004-02-01T00:00:00.000000000, number in-situ = 376

Year = 2005-02-01T00:00:00.000000000, number in-situ = 378

Year = 2006-02-01T00:00:00.000000000, number in-situ = 385

Year = 2007-02-01T00:00:00.000000000, number in-situ = 390

Year = 2008-02-01T00:00:00.000000000, number in-situ = 390

Year = 2009-02-01T00:00:00.000000000, number in-situ = 398

Year = 2010-02-01T00:00:00.000000000, number in-situ = 386

Year = 2011-02-01T00:00:00.000000000, number in-situ = 386

Year = 2012-02-01T00:00:00.000000000, number in-situ = 371

Year = 2013-02-01T00:00:00.000000000, number in-situ = 377

Year = 2014-02-01T00:00:00.000000000, number in-situ = 381

Year = 2015-02-01T00:00:00.000000000, number in-situ = 380

Year = 2016-02-01T00:00:00.000000000, number in-situ = 384

Year = 2017-02-01T00:00:00.000000000, number in-situ = 395

Year = 2018-02-01T00:00:00.000000000, number in-situ = 393

Year = 2019-02-01T00:00:00.000000000, number in-situ = 394

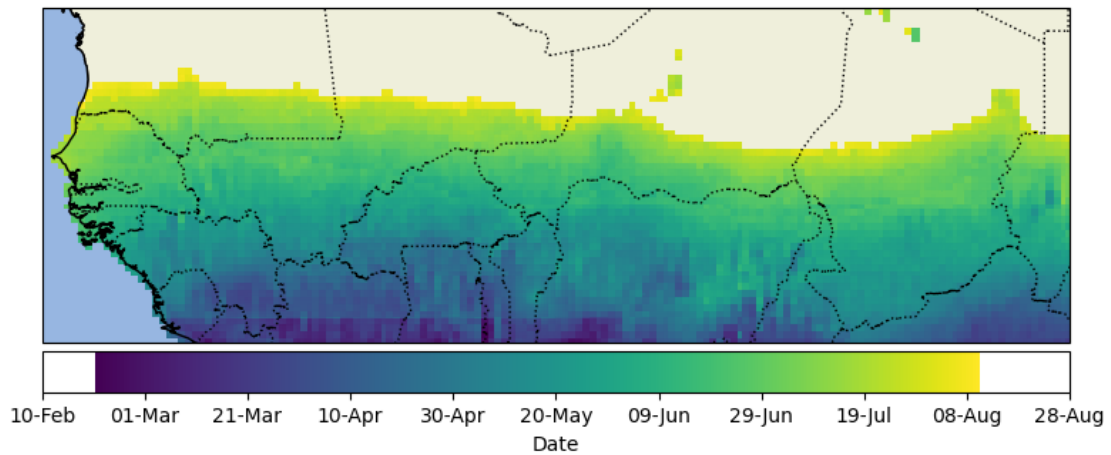
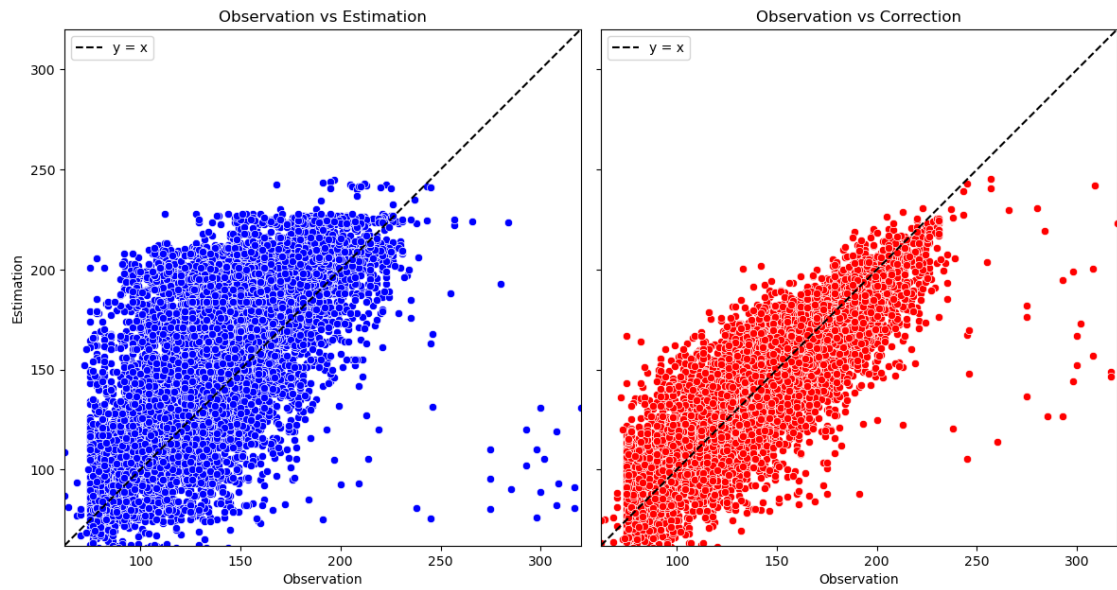
Year = 2020-02-01T00:00:00.000000000, number in-situ = 386

Year = 2021-02-01T00:00:00.000000000, number in-situ = 376

Year = 2022-02-01T00:00:00.000000000, number in-situ = 377

Year = 2023-02-01T00:00:00.000000000, number in-situ = 353

Year = 2024-02-01T00:00:00.000000000, number in-situ = 344



1 1st - Approach: Choice of Best Dynamical models and weighted Average

Agro_PRESASS_2025/daily_model_data/hindcast_ecmwf51_PRCP_Mar01_1993_2016_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/hindcast_meteo_france8_PRCP_Mar01_1993_2016_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/hindcast_ukmo604_PRCP_Mar01_1993_2016_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/hindcast_dwd21_PRCP_Mar01_1993_2016_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/hindcast_dwd22_PRCP_Mar01_1993_2016_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/hindcast_cmcc35_PRCP_Mar01_1993_2016_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/hindcast_eccc4_PRCP_Mar01_1993_2016_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/hindcast_eccc5_PRCP_Mar01_1993_2016_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/hindcast_jma3_PRCP_Mar01_1993_2016_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/hindcast_ncep2_PRCP_Mar01_1993_2016_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/forecast_ecmwf51_PRCP_Mar01_2025_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/forecast_meteo_france8_PRCP_Mar01_2025_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/forecast_ukmo604_PRCP_Mar01_2025_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/forecast_dwd21_PRCP_Mar01_2025_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/forecast_dwd22_PRCP_Mar01_2025_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/forecast_cmcc35_PRCP_Mar01_2025_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/forecast_eccc4_PRCP_Mar01_2025_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/forecast_eccc5_PRCP_Mar01_2025_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/forecast_jma3_PRCP_Mar01_2025_24-5160.nc already exists. Skipping download.

Agro_PRESASS_2025/daily_model_data/forecast_ncep2_PRCP_Mar01_2025_24-5160.nc already exists. Skipping download.

[SKIP] Agro_PRESASS_2025/model_data/hindcast_ecmwf51_Onset_MarIc.nc already exists.

[SKIP] Agro_PRESASS_2025/model_data/hindcast_meteo_france8_Onset_MarIc.nc already exists.

[SKIP] Agro_PRESASS_2025/model_data/hindcast_ukmo604_Onset_MarIc.nc already exists.

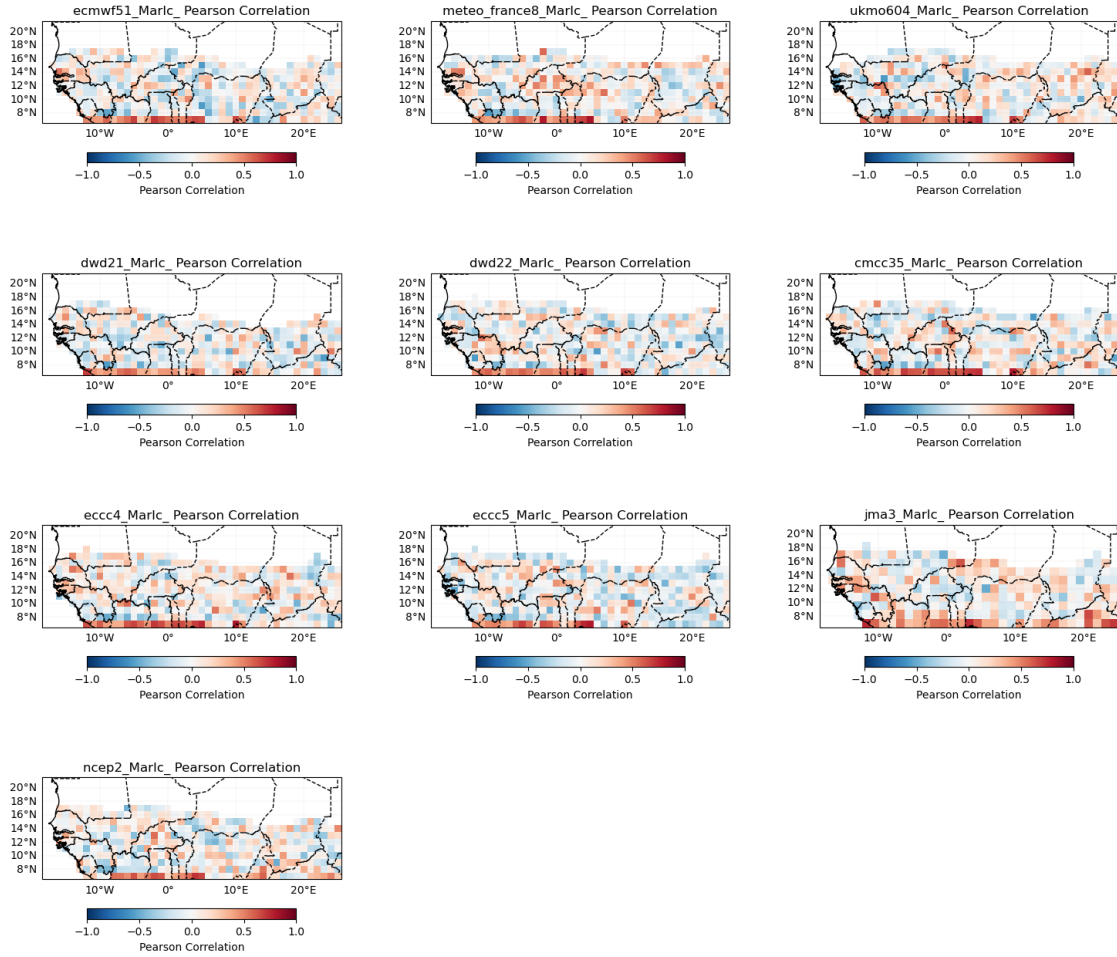
[SKIP] Agro_PRESASS_2025/model_data/hindcast_dwd21_Onset_MarIc.nc already exists.

[SKIP] Agro_PRESASS_2025/model_data/hindcast_dwd22_Onset_MarIc.nc already exists.

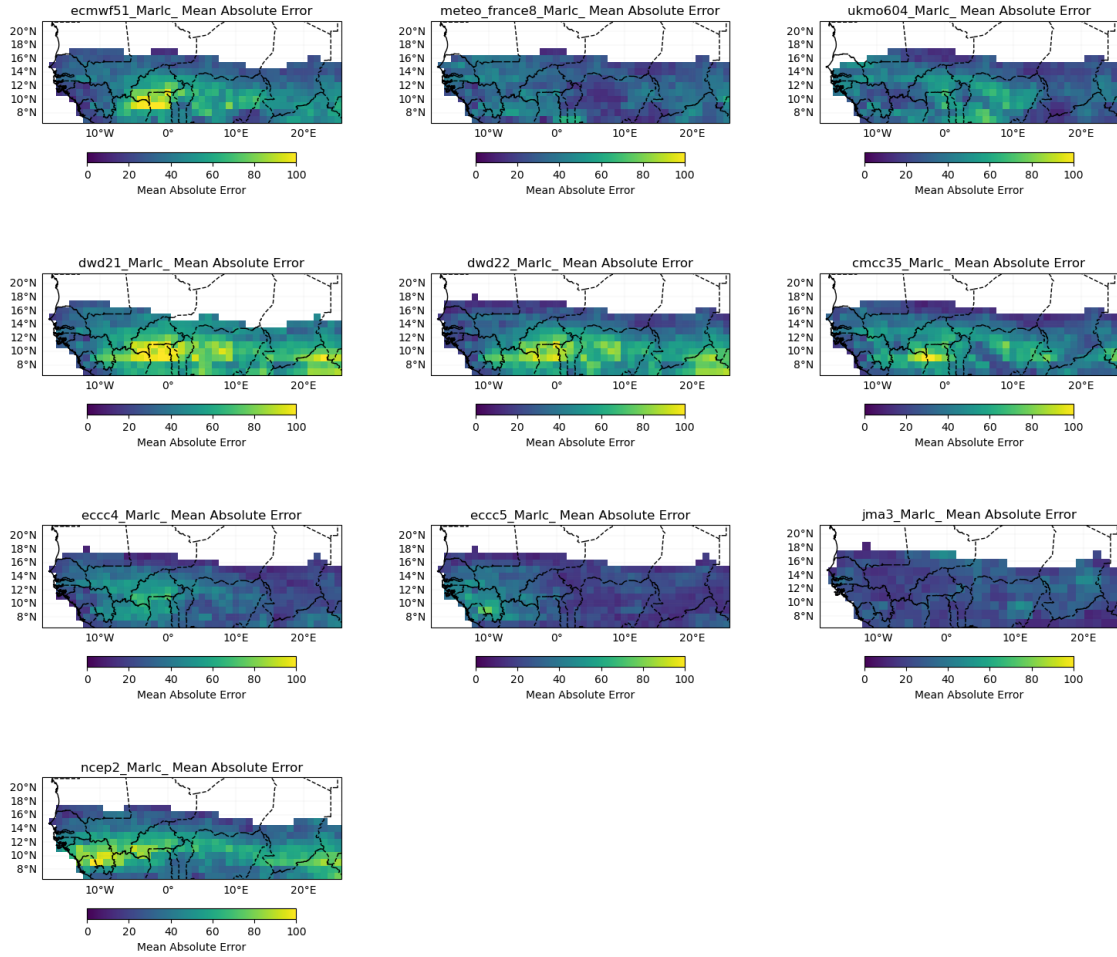
exists.
[SKIP] Agro_PRESASS_2025/model_data/hindcast_cmcc35_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/hindcast_eccc4_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/hindcast_eccc5_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/hindcast_jma3_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/hindcast_ncep2_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/forecast_ecmwf51_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/forecast_meteo_france8_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/forecast_ukmo604_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/forecast_dwd21_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/forecast_dwd22_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/forecast_cmcc35_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/forecast_eccc4_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/forecast_eccc5_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/forecast_jma3_Onset_MarIc.nc already exists.
[SKIP] Agro_PRESASS_2025/model_data/forecast_ncep2_Onset_MarIc.nc already exists.

Validation of GCM

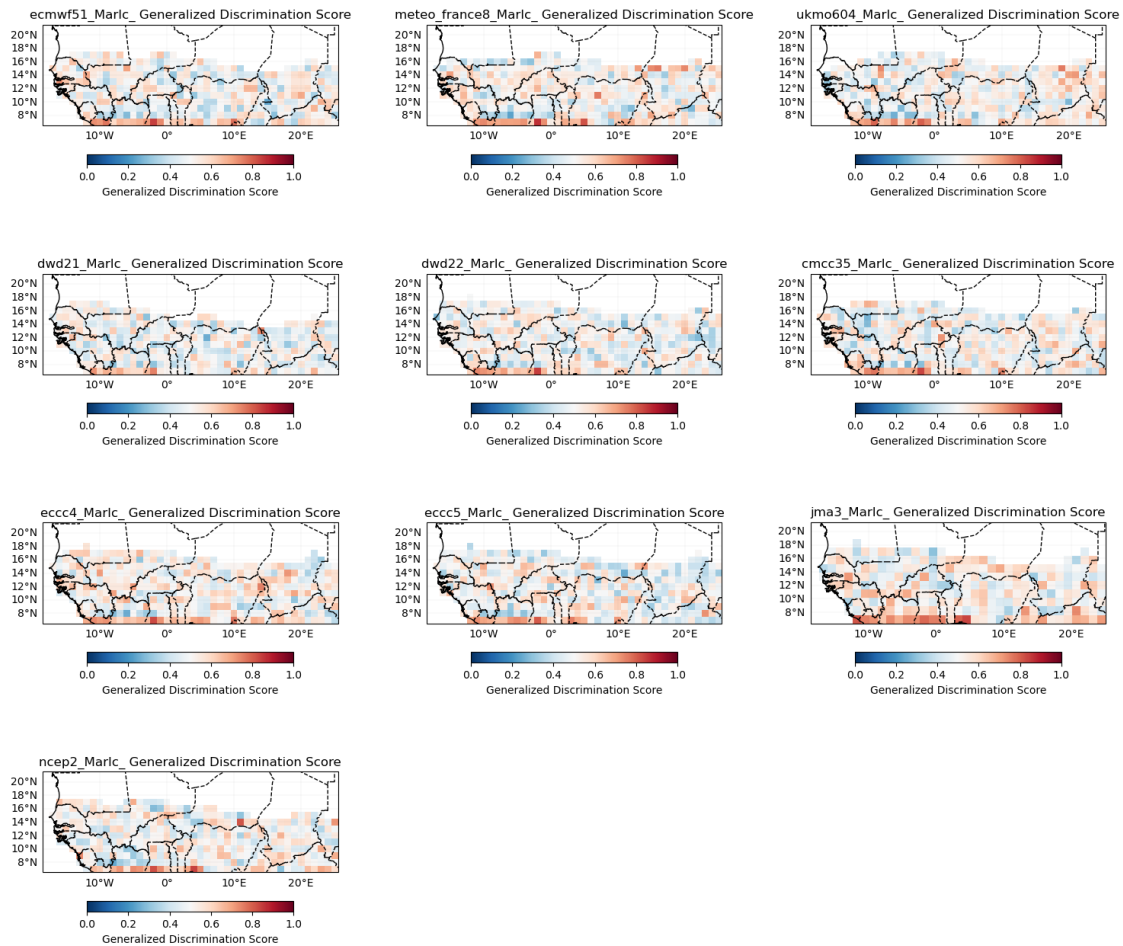
Pearson Correlation



Mean Absolute Error



Generalized Discrimination Score



1.1 Choice of best GCM models

`['ECCC_4.PRCP', 'METEOFRACTANCE_8.PRCP', 'UKMO_604.PRCP']`

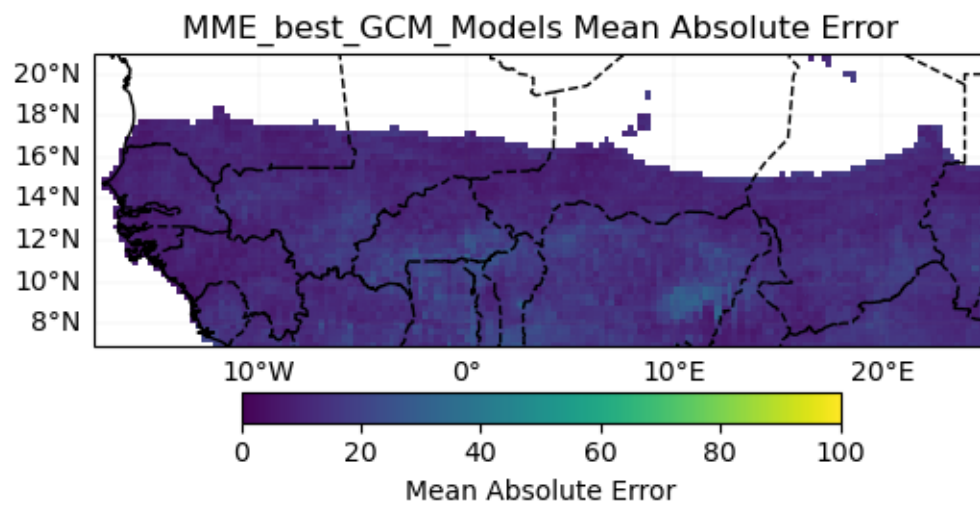
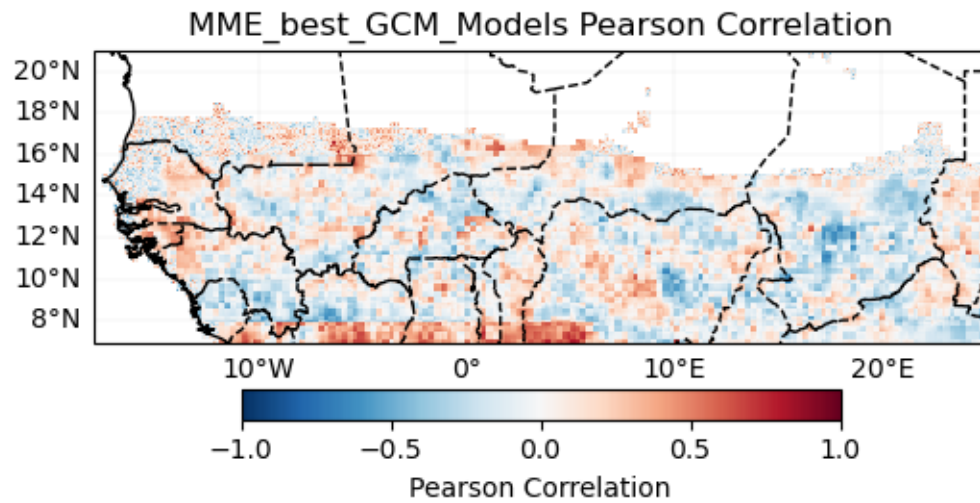
1.2 MME for best GCM models

Cross-validation

Cross-validation ongoing

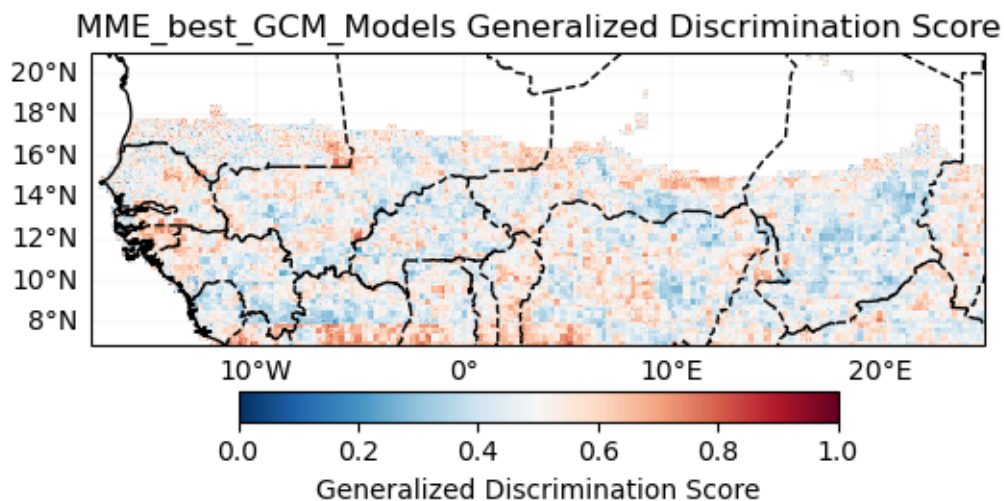
100% | 24/24 [03:05<00:00, 7.75s/it]

Verification

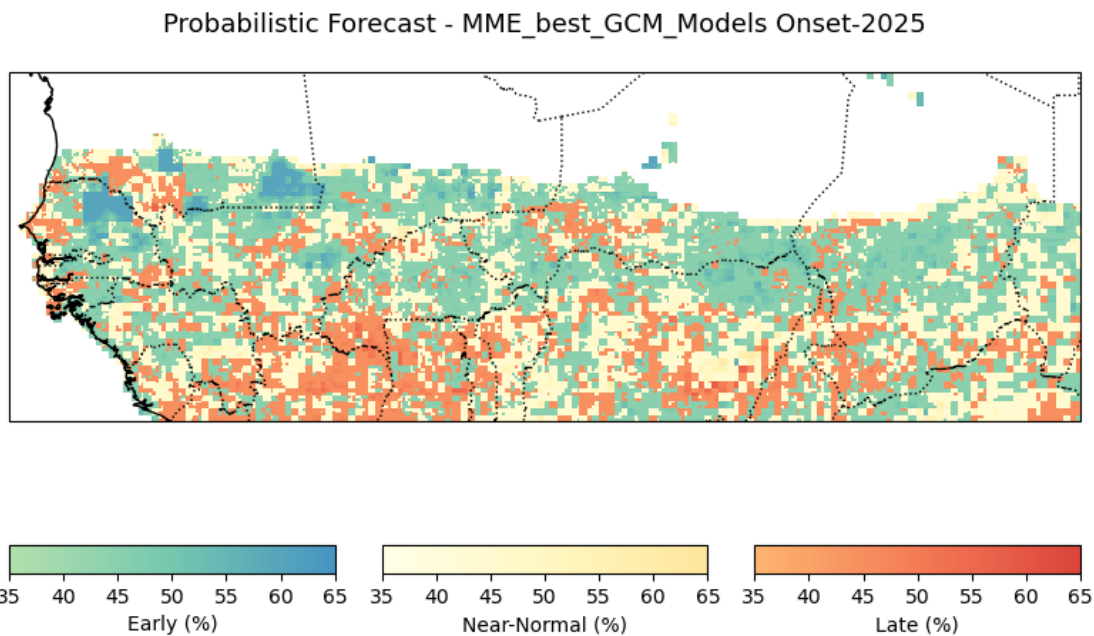


Probabilistic

Generalized Discrimination Score



Forecast



119974

2 2nd - Approach: Statistico-Dynamical

Agro_PRESASS_2025/model_data/hindcast_ecmwf51_VGRD_850_AprIc_MayJunJul_01.nc already exists. Skipping download.

Agro_PRESASS_2025/model_data/hindcast_meteofrance8_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/hindcast_ukmo604_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/hindcast_dwd22_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/hindcast_cmcc35_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/hindcast_eccc4_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/hindcast_eccc5_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/hindcast_jma3_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/hindcast_ncep2_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/forecast_ecmwf51_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/forecast_meteofrance8_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/forecast_ukmo604_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/forecast_dwd22_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/forecast_cmcc35_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

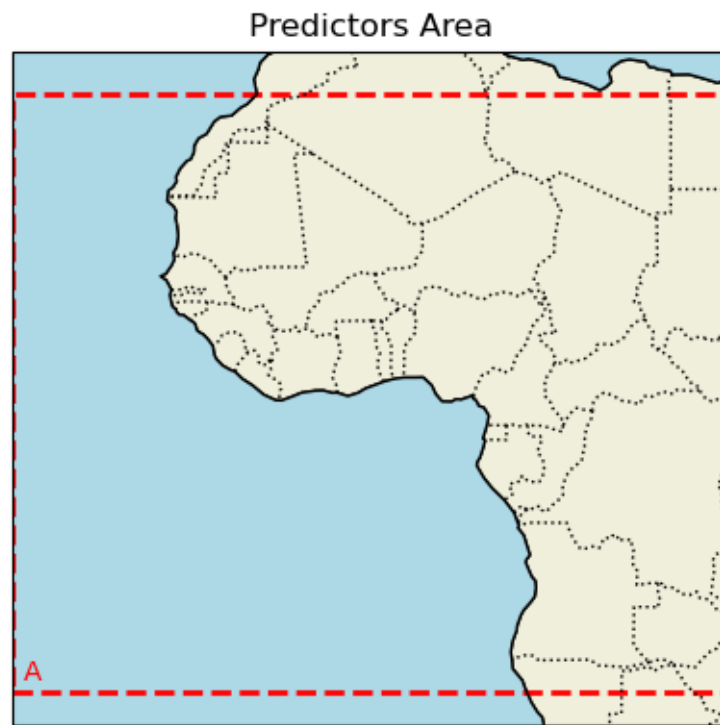
Agro_PRESASS_2025/model_data/forecast_eccc4_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/forecast_eccc5_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

Agro_PRESASS_2025/model_data/forecast_jma3_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

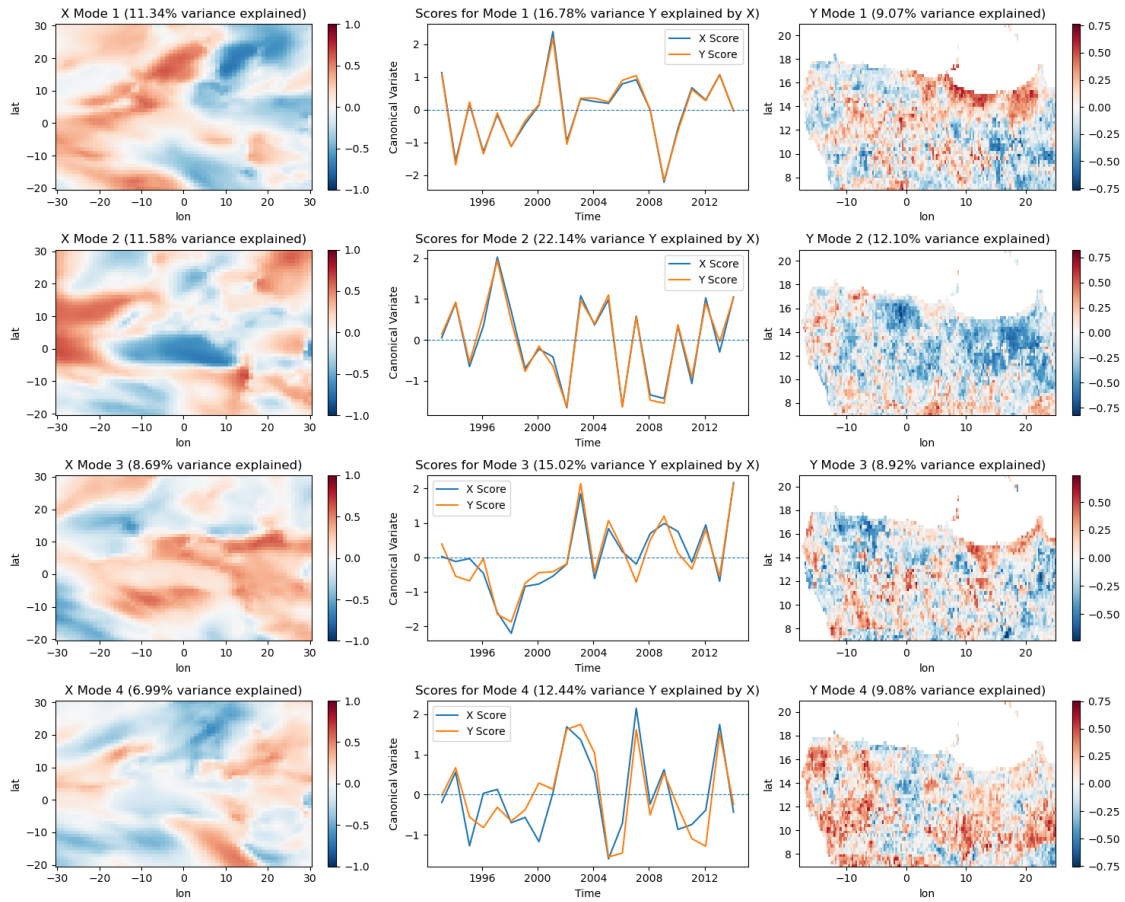
Agro_PRESASS_2025/model_data/forecast_ncep2_VGRD_850_AprIc_MayJunJul_01.nc
already exists. Skipping download.

2.0.1 Initialize the WAS_CCA class (default)

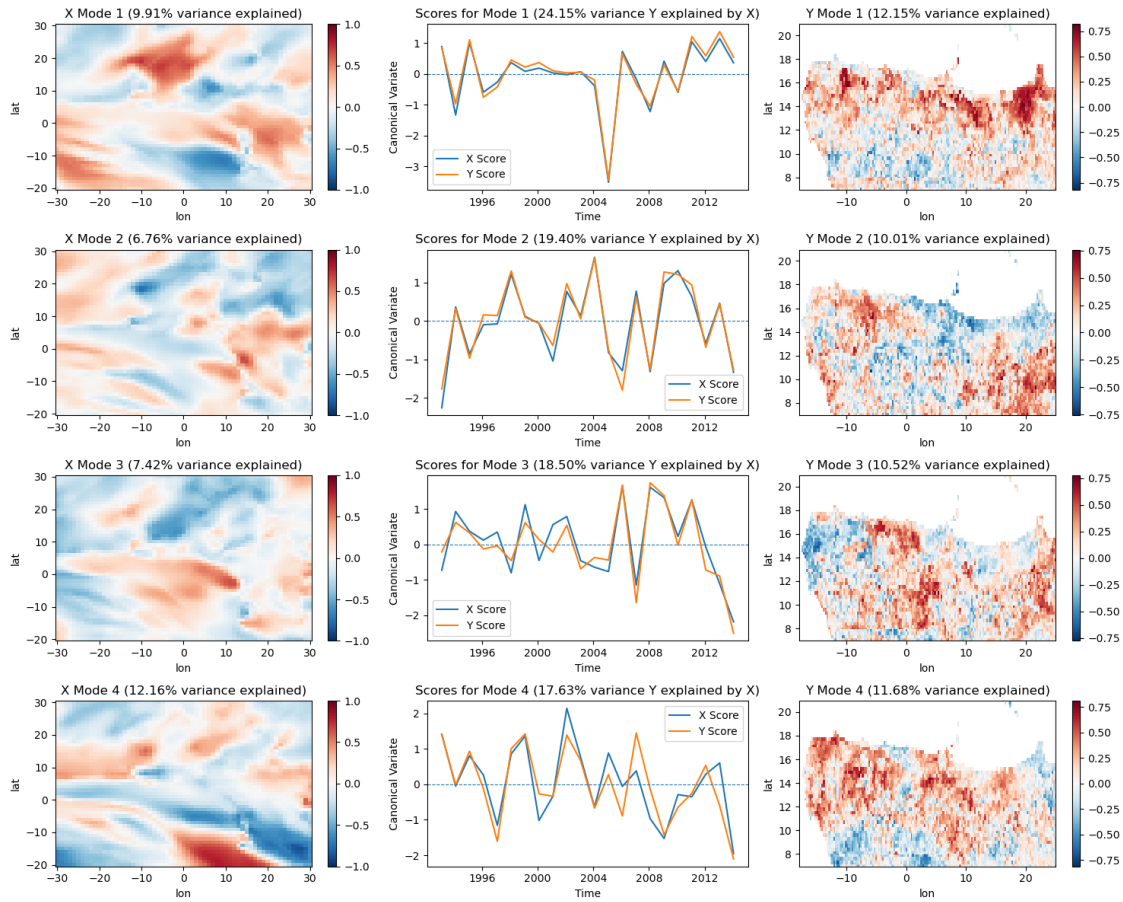


Modes of CCA for each model

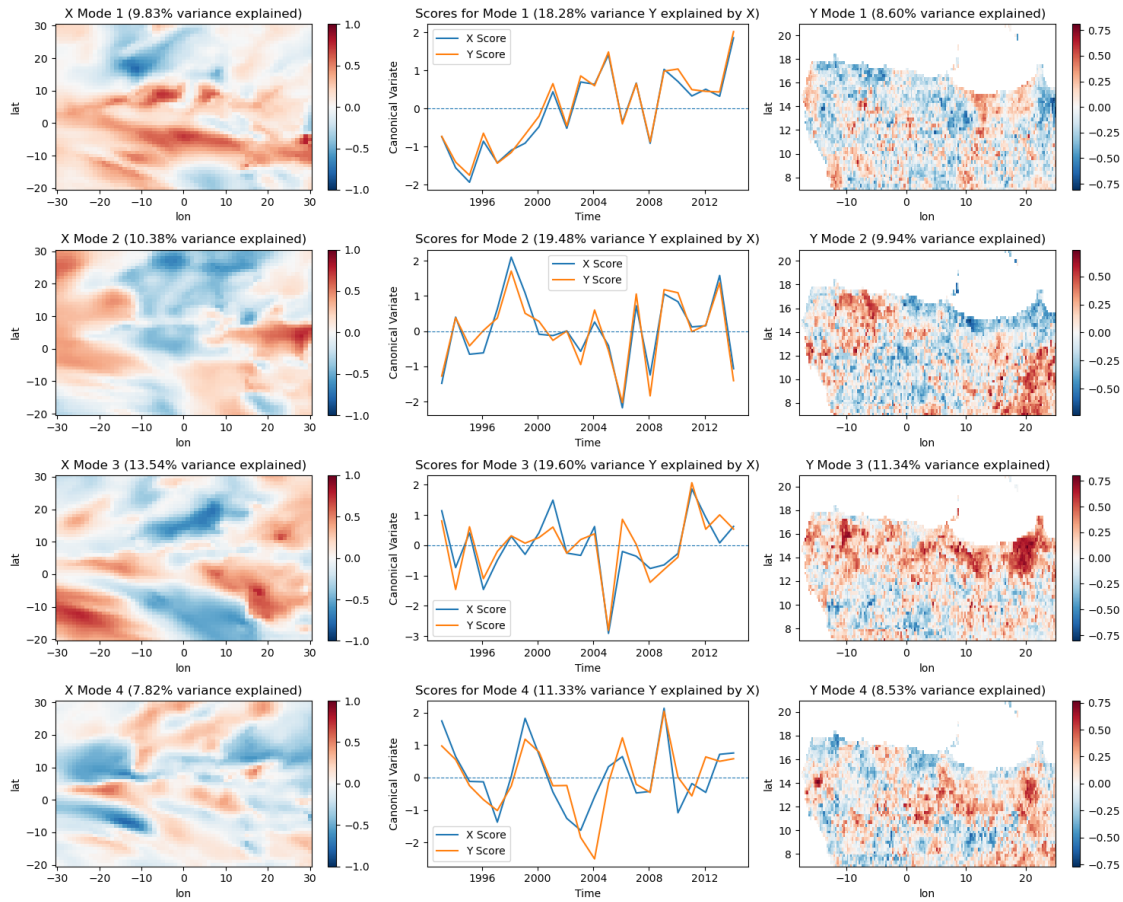
CCA Modes for model ECMWF_51.VGRD_850



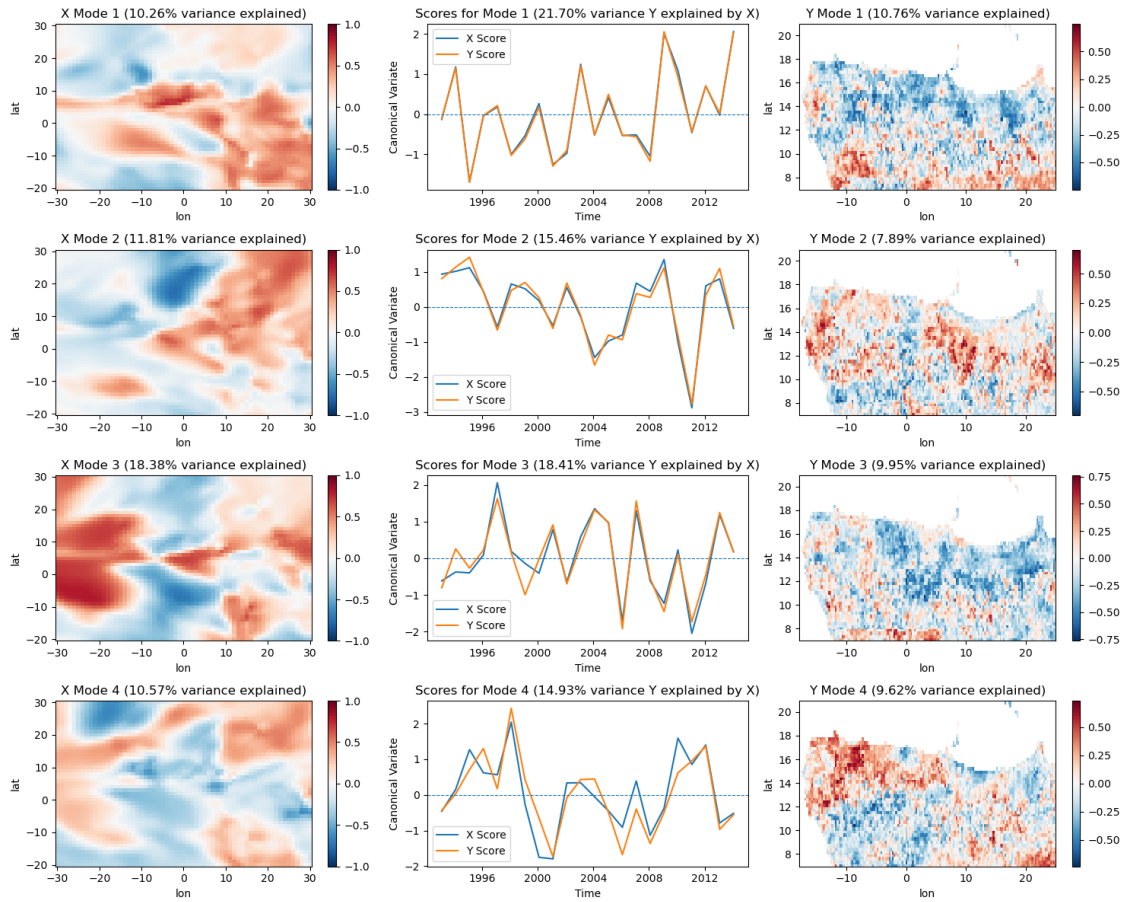
CCA Modes for model METEOFRACTANCE_8.VGRD_850



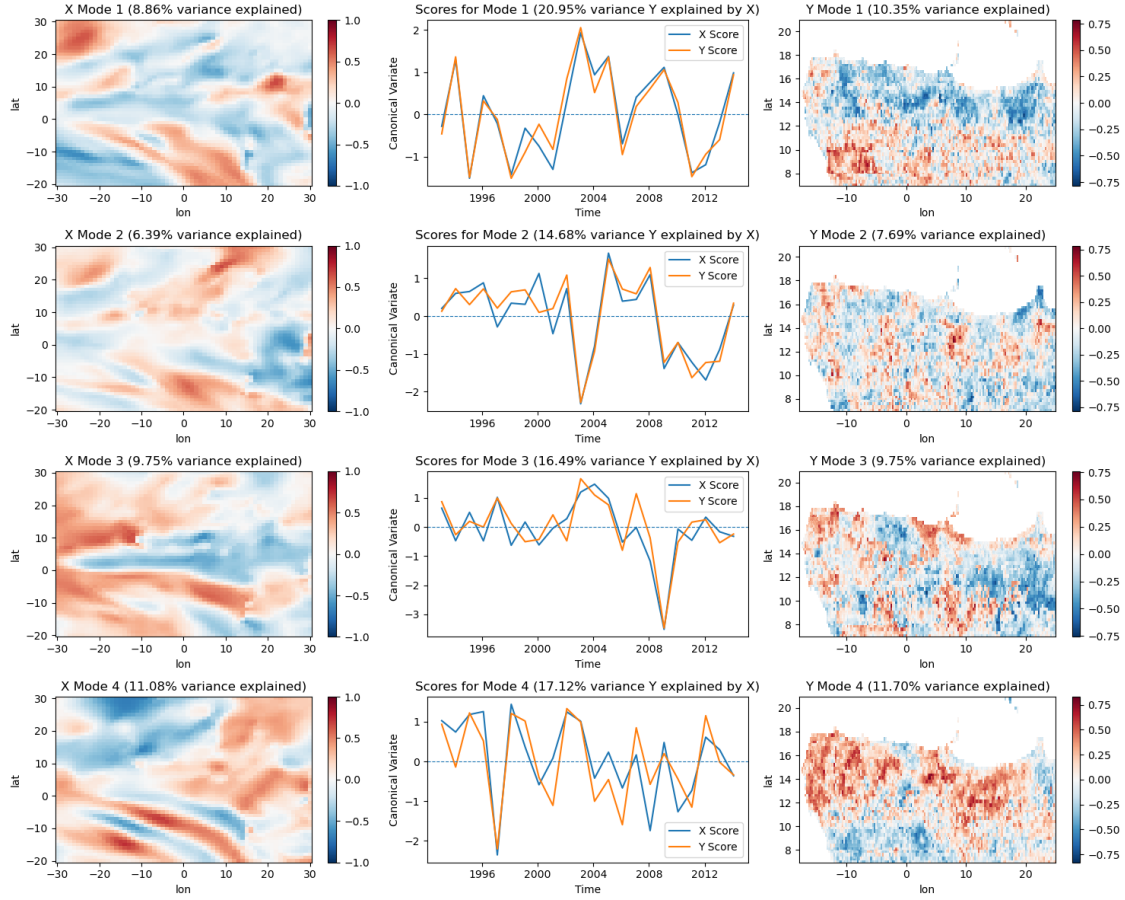
CCA Modes for model UKMO_604.VGRD_850



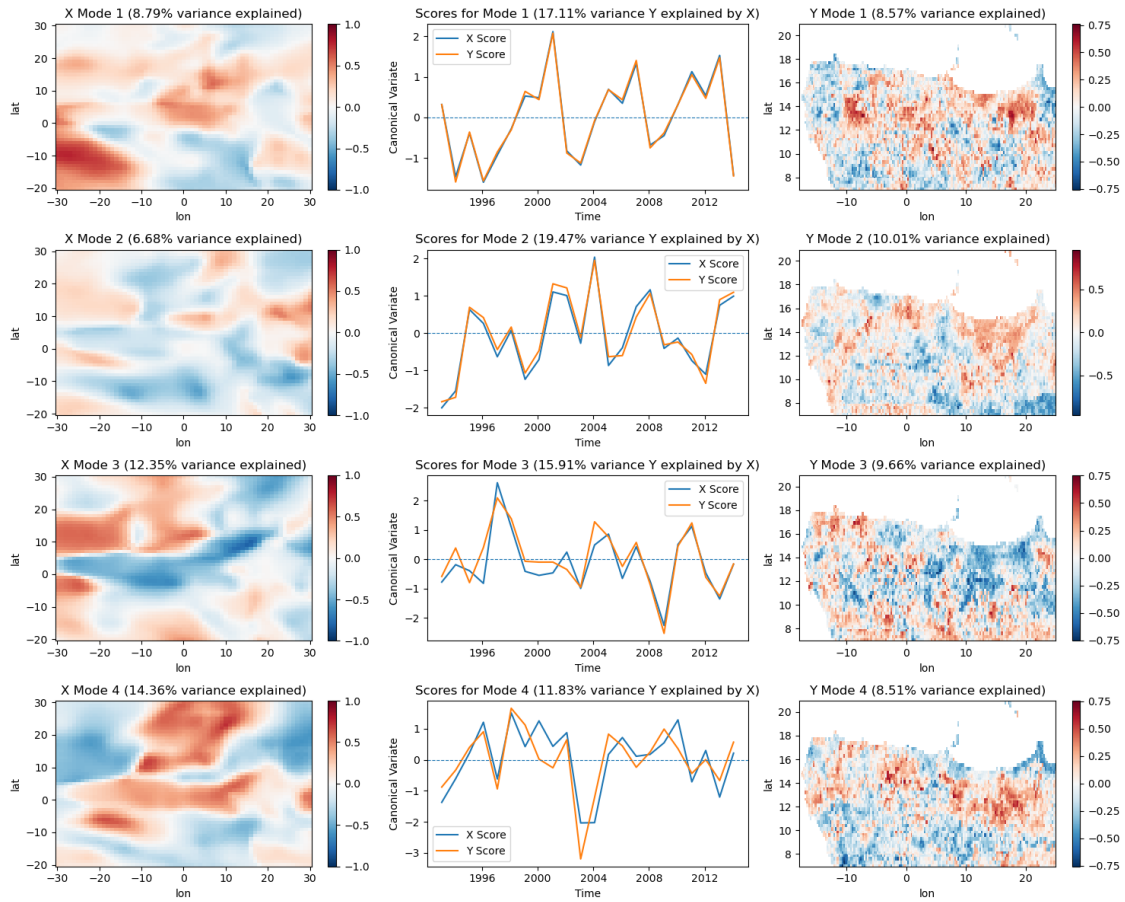
CCA Modes for model DWD_22.VGRD_850



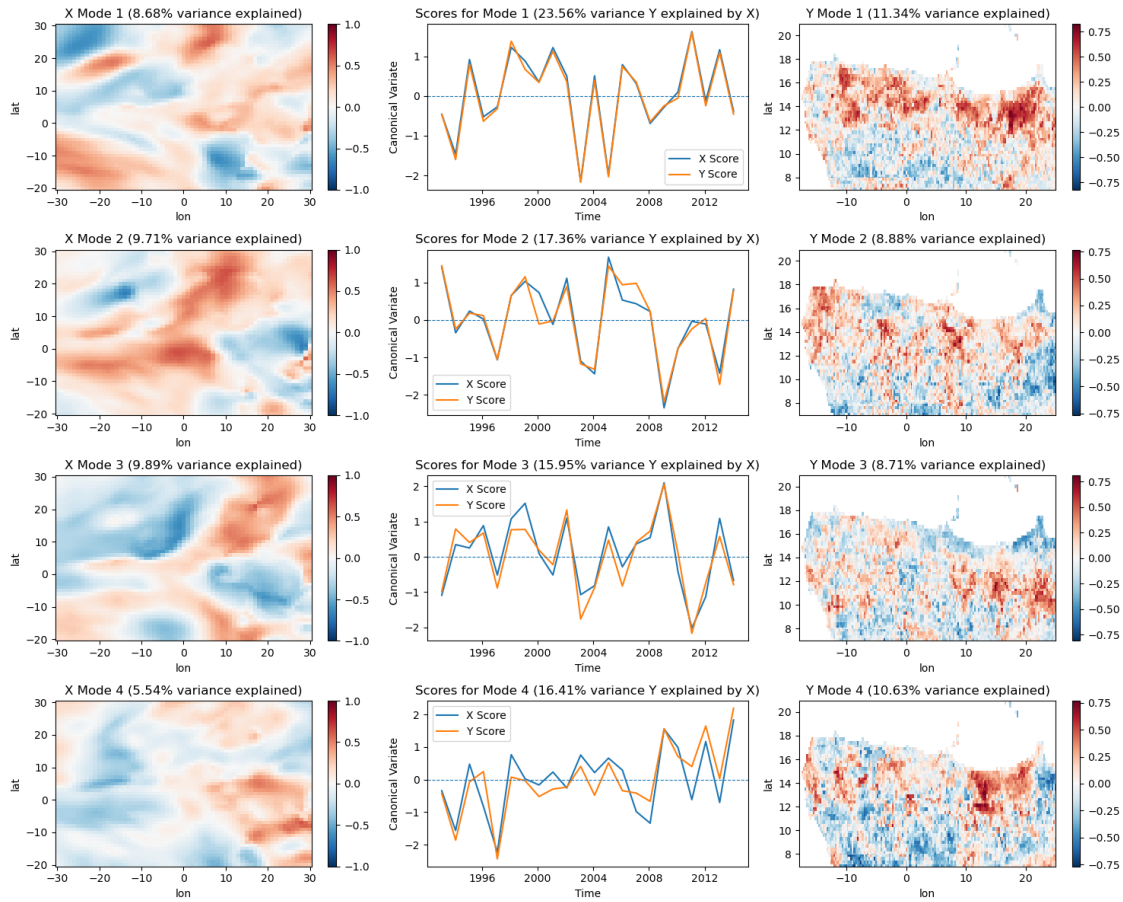
CCA Modes for model CMCC_35.VGRD_850



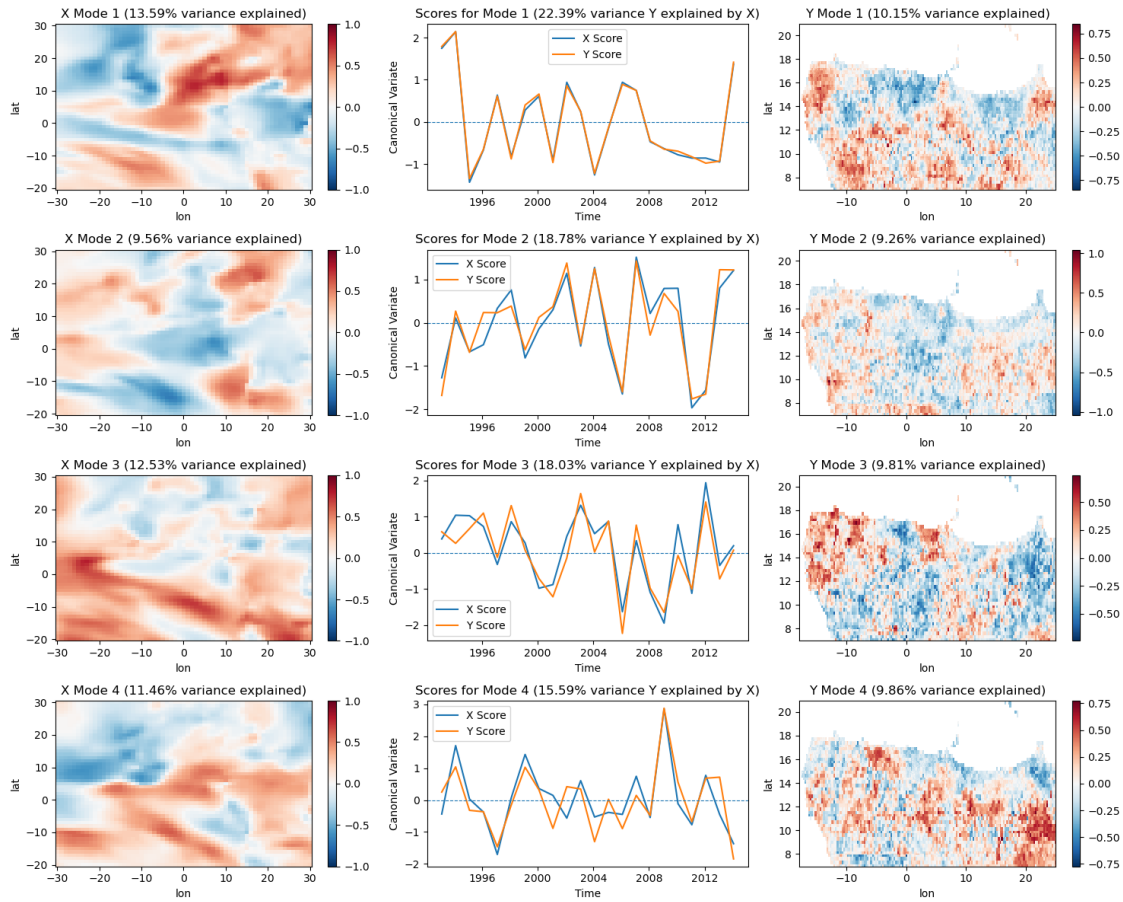
CCA Modes for model ECCC_4.VGRD_850



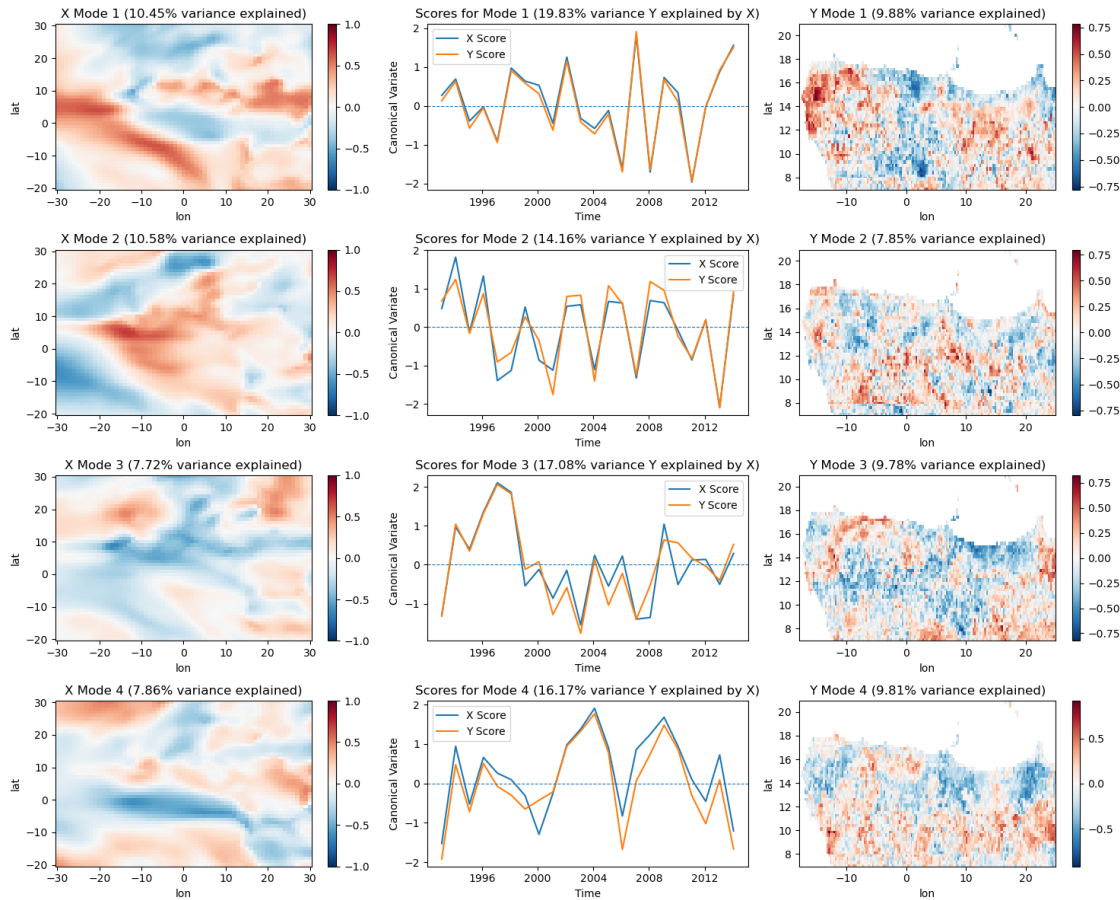
CCA Modes for model ECCC_5.VGRD_850



CCA Modes for model JMA_3.VGRD_850



CCA Modes for model NCEP_2.VGRD_850



Cross validation

ECMWF_51

Cross-validation ongoing

100% | 24/24 [00:06<00:00, 3.67it/s]

METEOFRANCE_8

Cross-validation ongoing

100% | 24/24 [00:06<00:00, 3.47it/s]

UKMO_604

Cross-validation ongoing

100% | 24/24 [00:07<00:00, 3.36it/s]

DWD_22

Cross-validation ongoing

100% | 24/24 [00:06<00:00, 3.45it/s]

CMCC_35

Cross-validation ongoing

100%| | 24/24 [00:07<00:00, 3.28it/s]

ECCC_4

Cross-validation ongoing

100%| | 24/24 [00:06<00:00, 3.69it/s]

ECCC_5

Cross-validation ongoing

100%| | 24/24 [00:06<00:00, 3.75it/s]

JMA_3

Cross-validation ongoing

100%| | 24/24 [00:06<00:00, 3.63it/s]

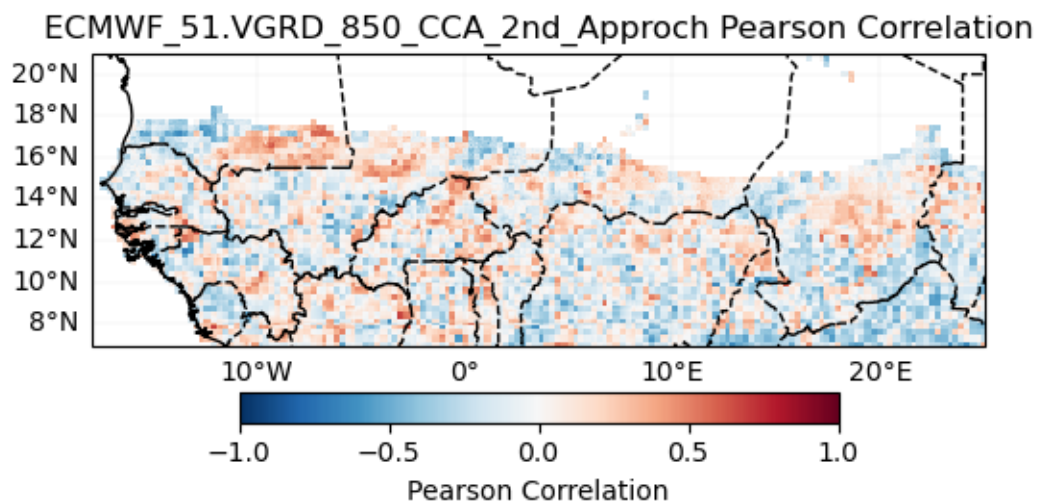
NCEP_2

Cross-validation ongoing

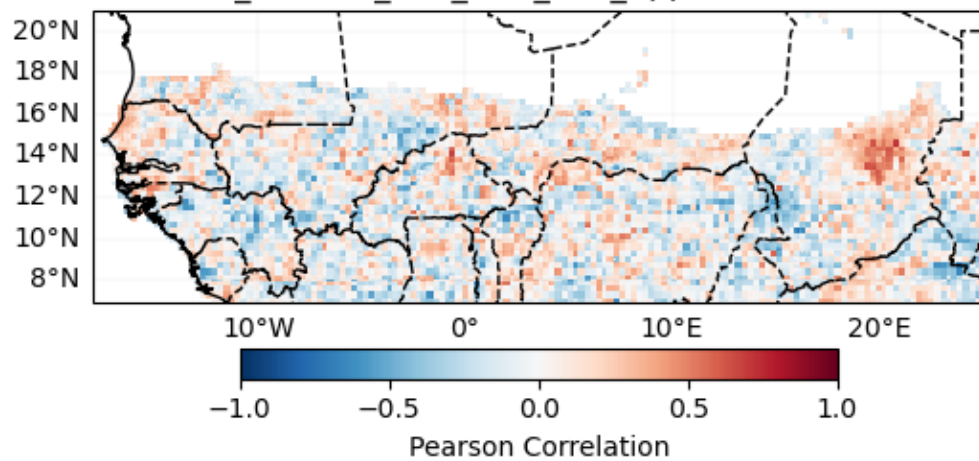
100%| | 24/24 [00:06<00:00, 3.74it/s]

Verification (individual) Deterministic

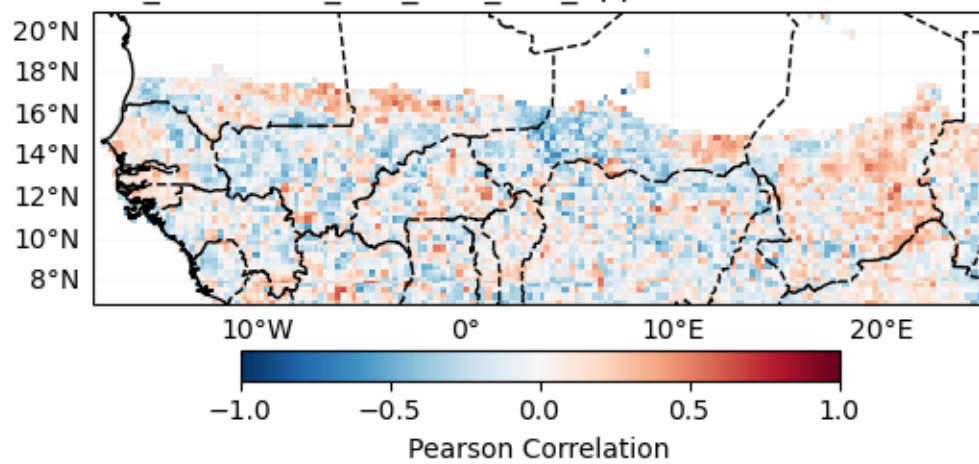
Pearson Correlation



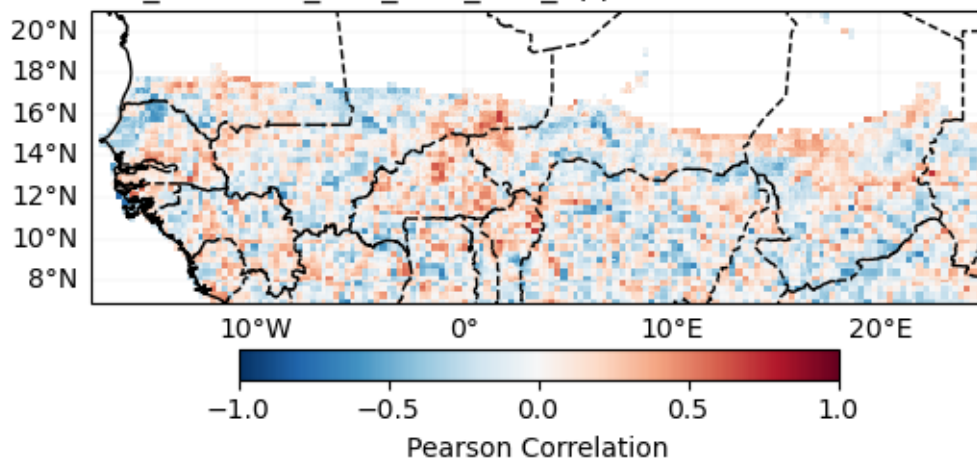
METEOFRACTANCE_8.VGRD_850_CCA_2nd_Approch Pearson Correlation



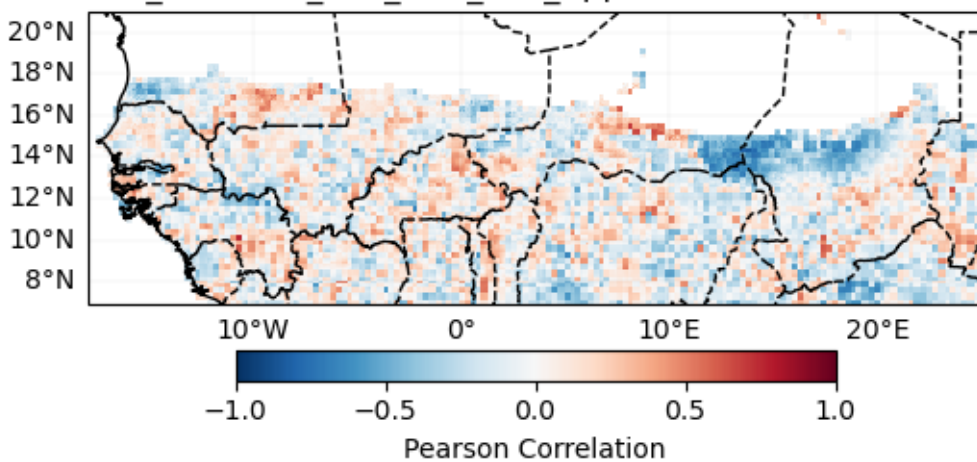
UKMO_604.VGRD_850_CCA_2nd_Approch Pearson Correlation



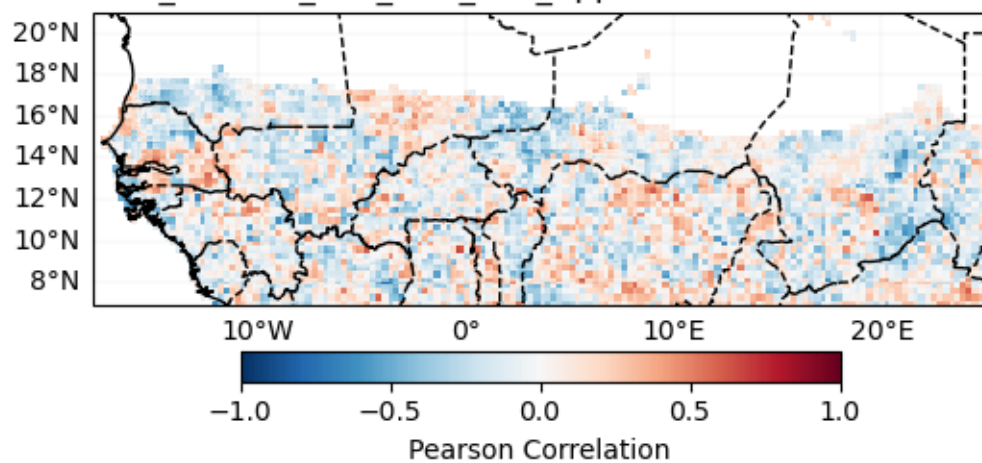
DWD_22.VGRD_850_CCA_2nd_Approch Pearson Correlation



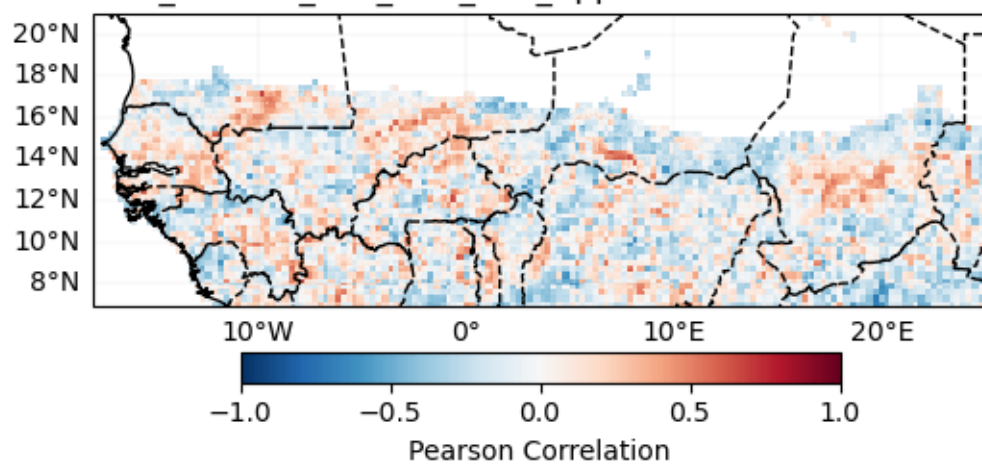
CMCC_35.VGRD_850_CCA_2nd_Approch Pearson Correlation



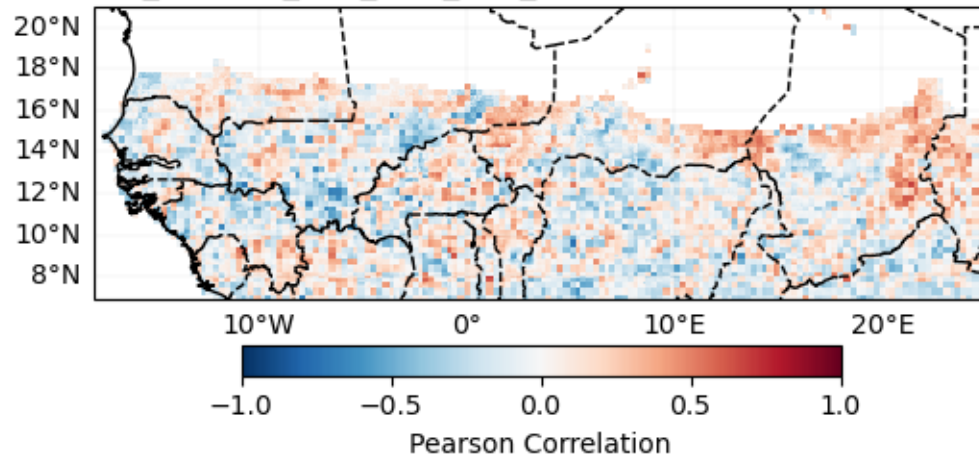
ECCC_4.VGRD_850_CCA_2nd_Approch Pearson Correlation



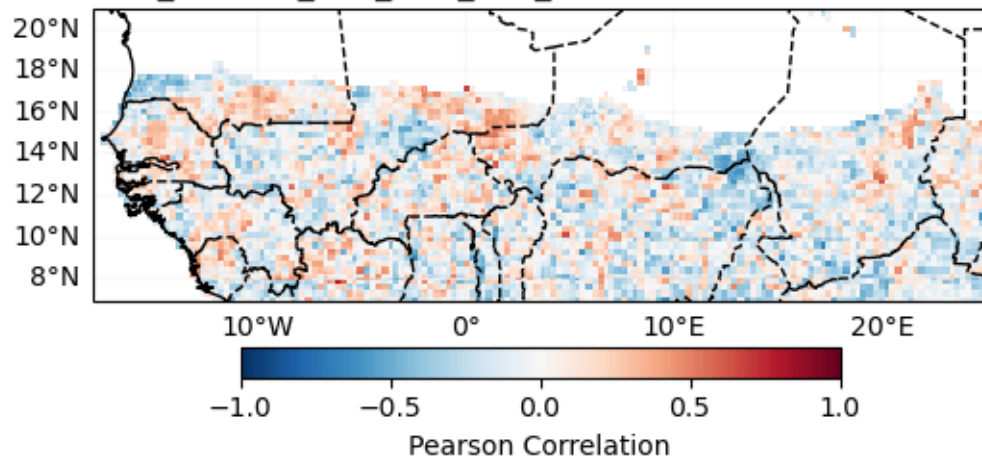
ECCC_5.VGRD_850_CCA_2nd_Approch Pearson Correlation



JMA_3.VGRD_850_CCA_2nd_Approch Pearson Correlation

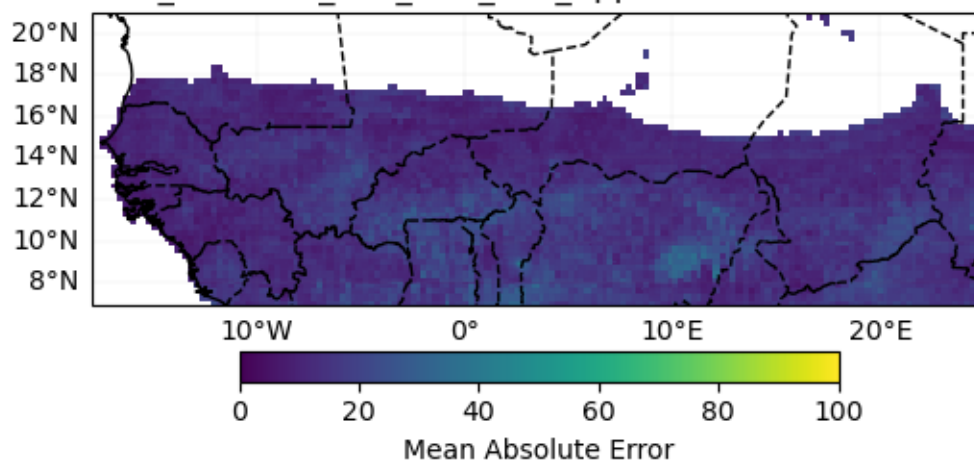


NCEP_2.VGRD_850_CCA_2nd_Approch Pearson Correlation

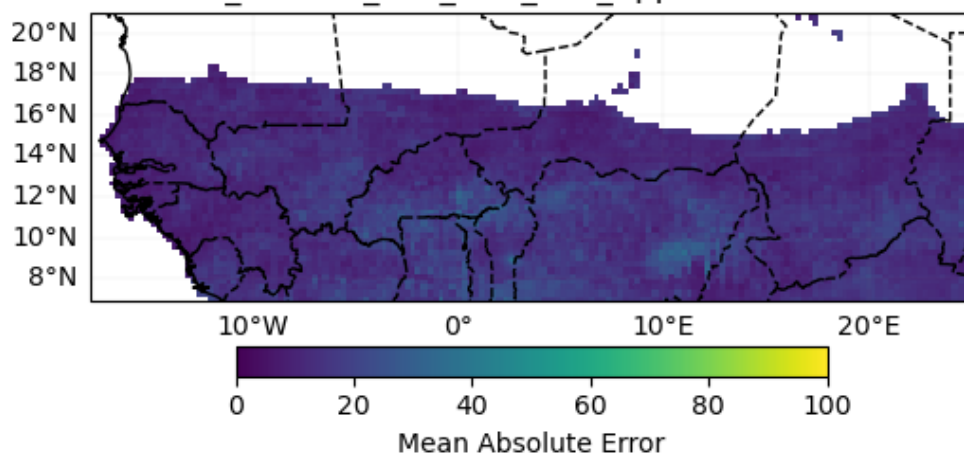


Mean Absolute Error

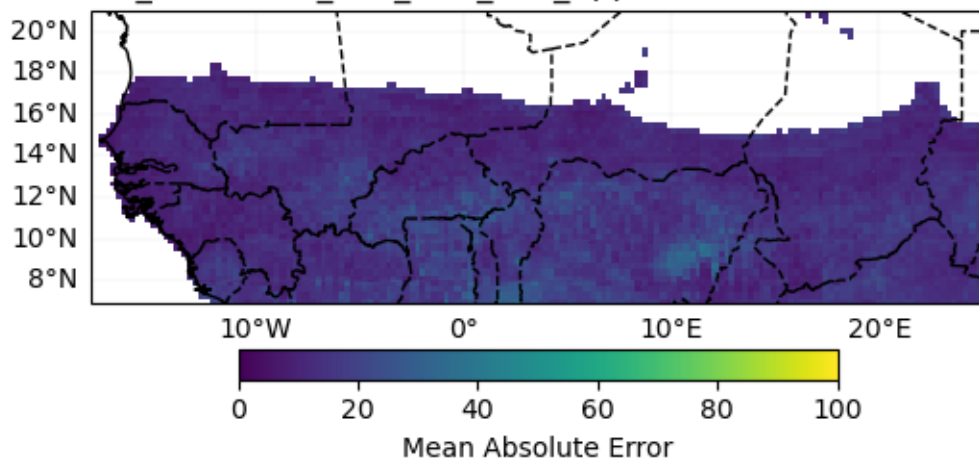
ECMWF_51.VGRD_850_CCA_2nd_Approch Mean Absolute Error



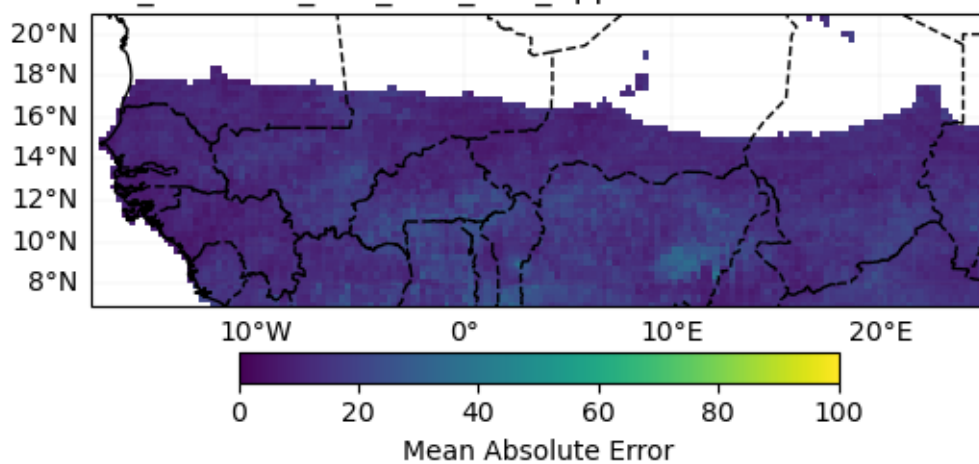
METEOFRANCE_8.VGRD_850_CCA_2nd_Approch Mean Absolute Error



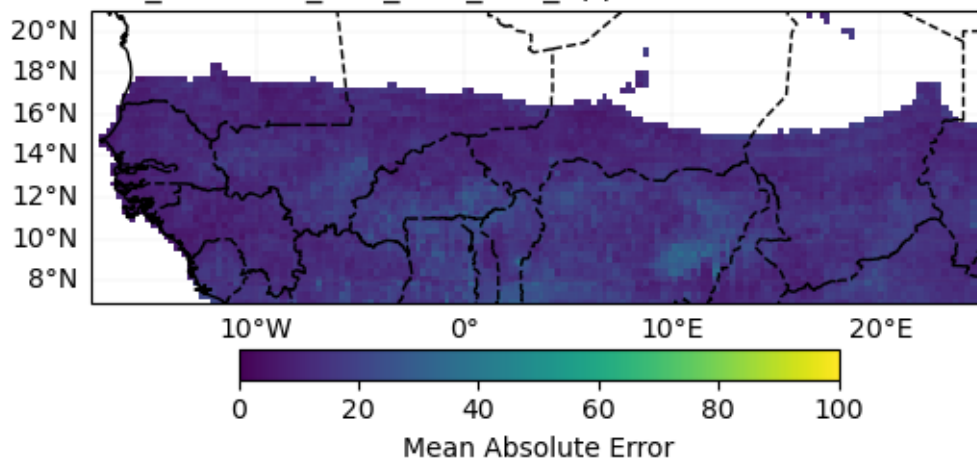
UKMO_604.VGRD_850_CCA_2nd_Approch Mean Absolute Error



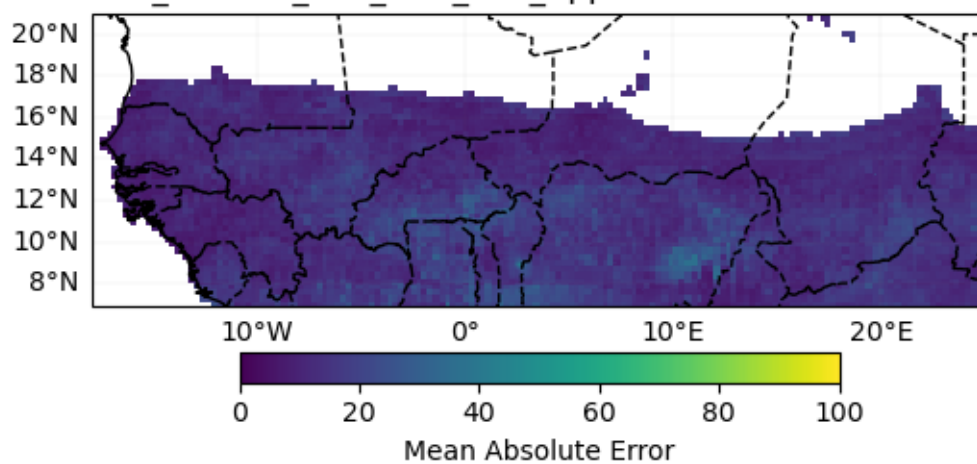
DWD_22.VGRD_850_CCA_2nd_Approch Mean Absolute Error



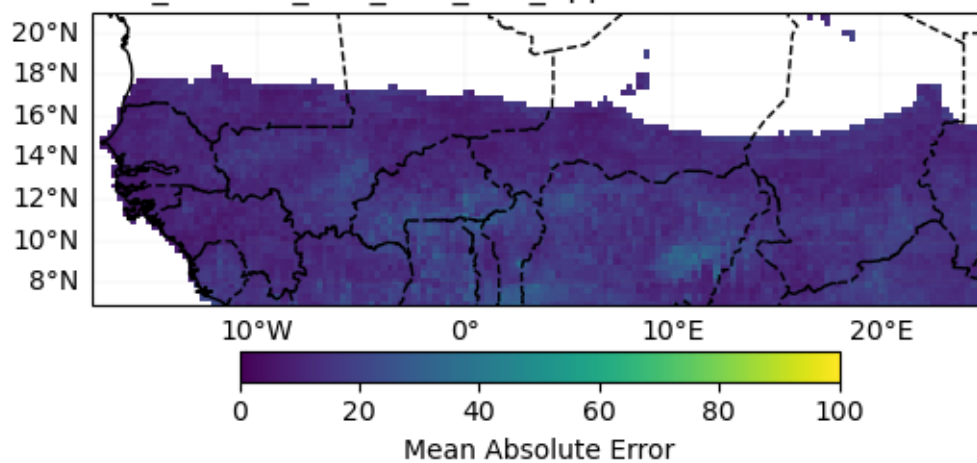
CMCC_35.VGRD_850_CCA_2nd_Approch Mean Absolute Error



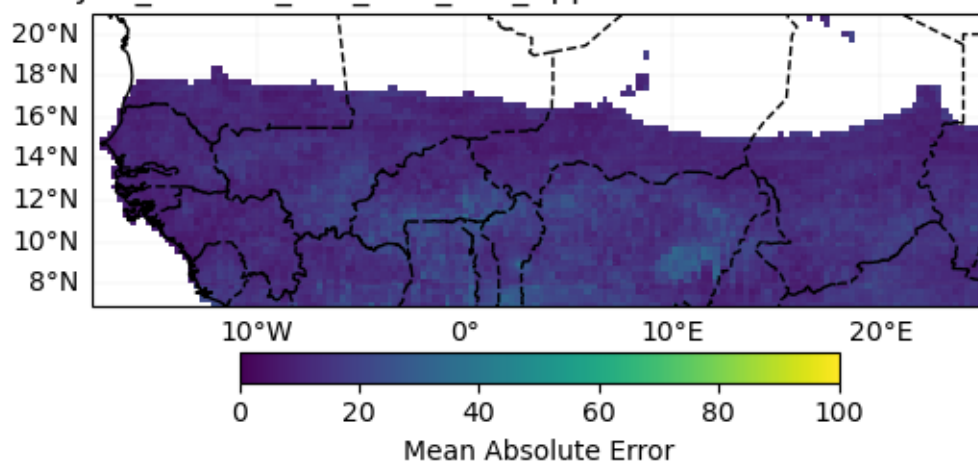
ECCC_4.VGRD_850_CCA_2nd_Approch Mean Absolute Error

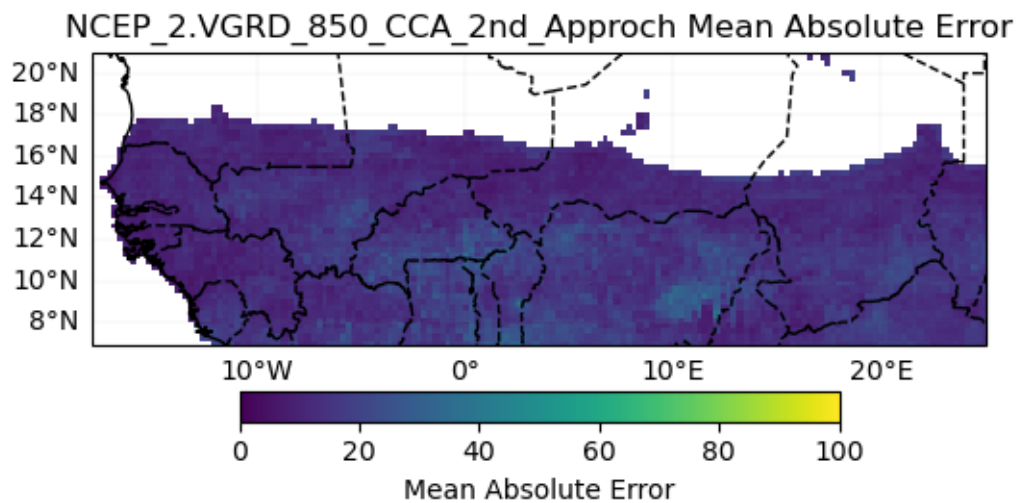


ECCC_5.VGRD_850_CCA_2nd_Approch Mean Absolute Error



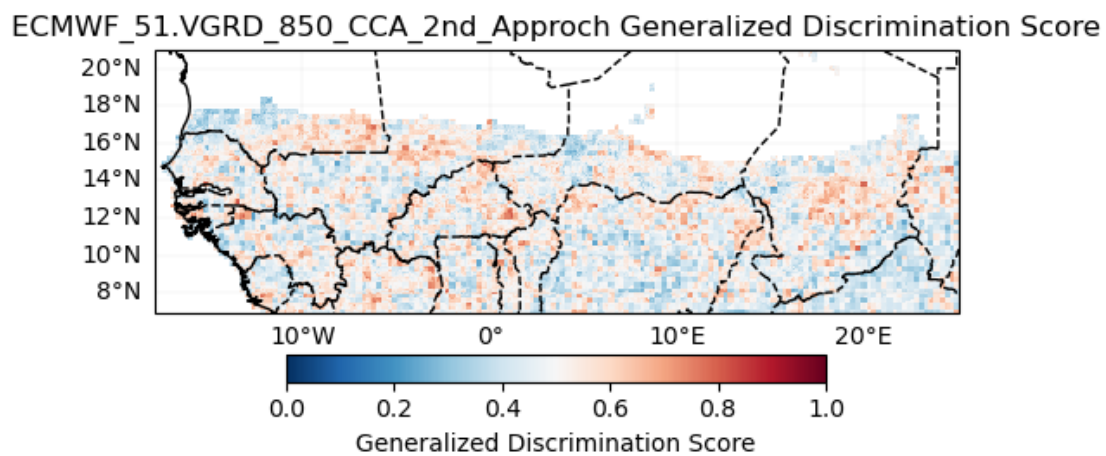
JMA_3.VGRD_850_CCA_2nd_Approch Mean Absolute Error



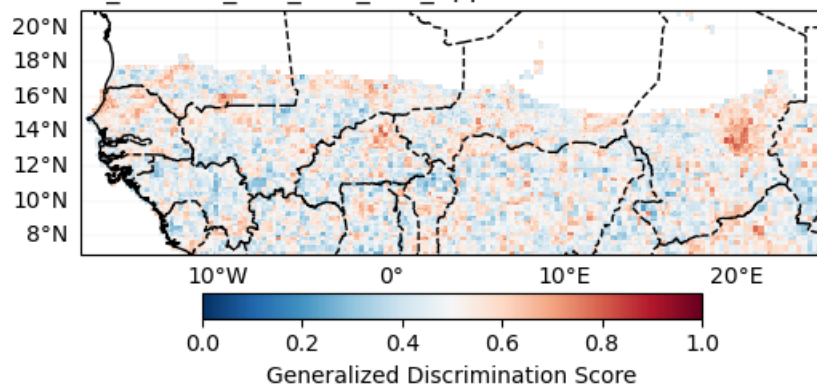


Probabilistic

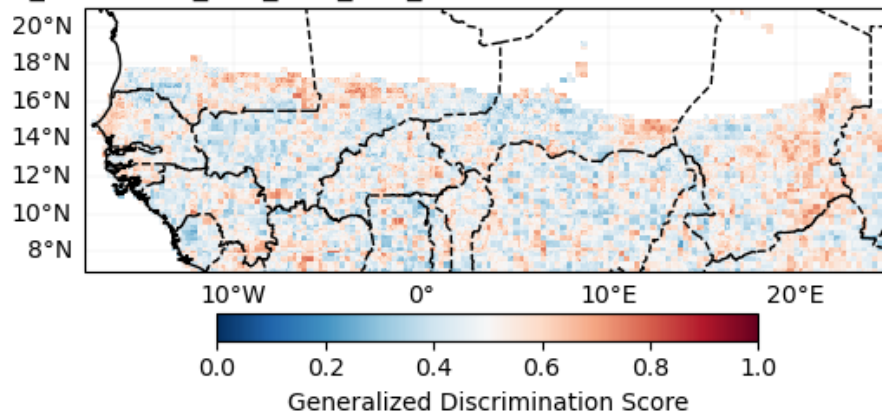
Generalized Discrimination Score



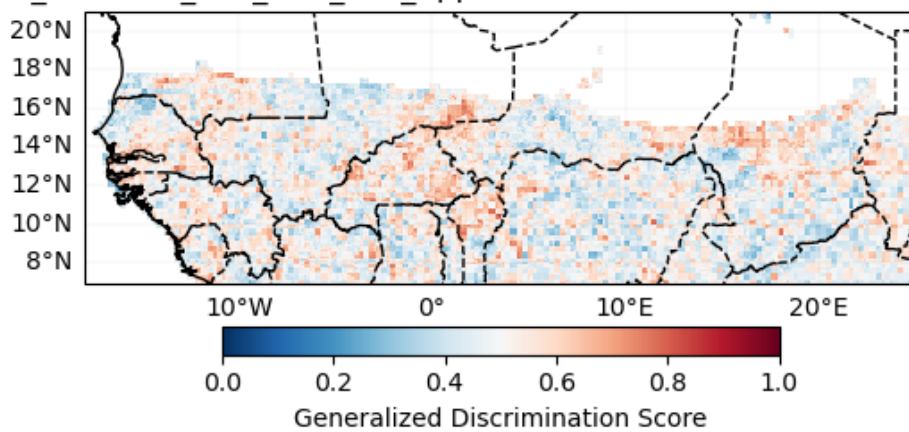
METEOFRANCE_8.VGRD_850_CCA_2nd_Approch Generalized Discrimination Score



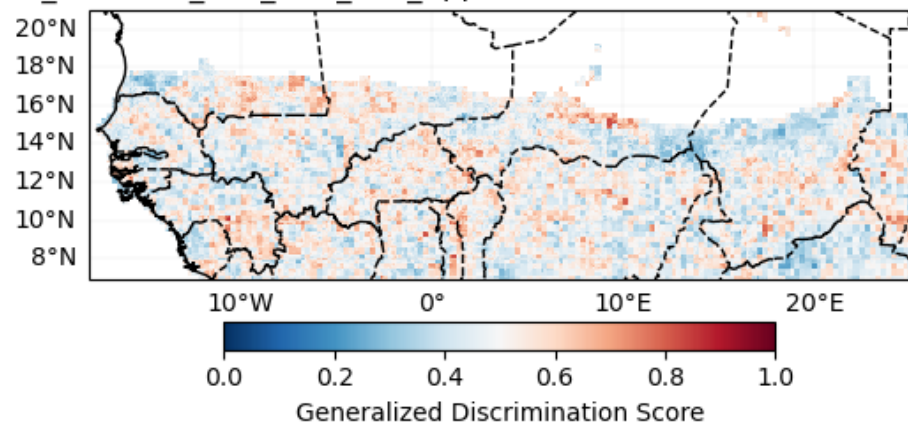
UKMO_604.VGRD_850_CCA_2nd_Approch Generalized Discrimination Score



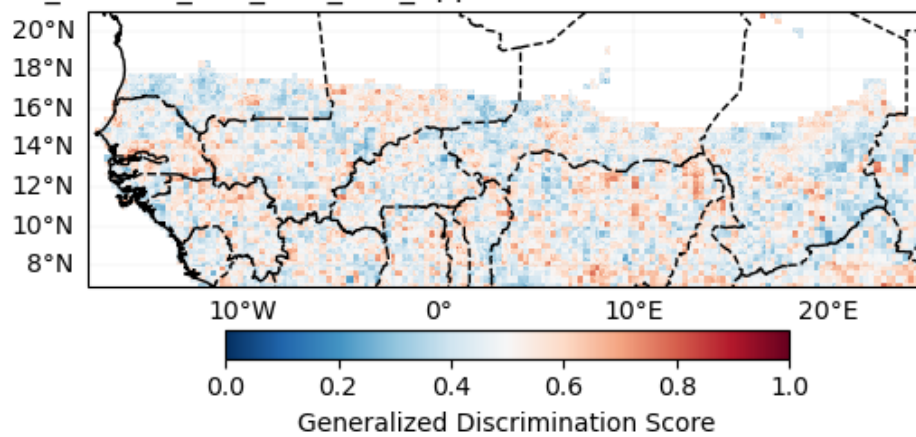
DWD_22.VGRD_850_CCA_2nd_Approch Generalized Discrimination Score



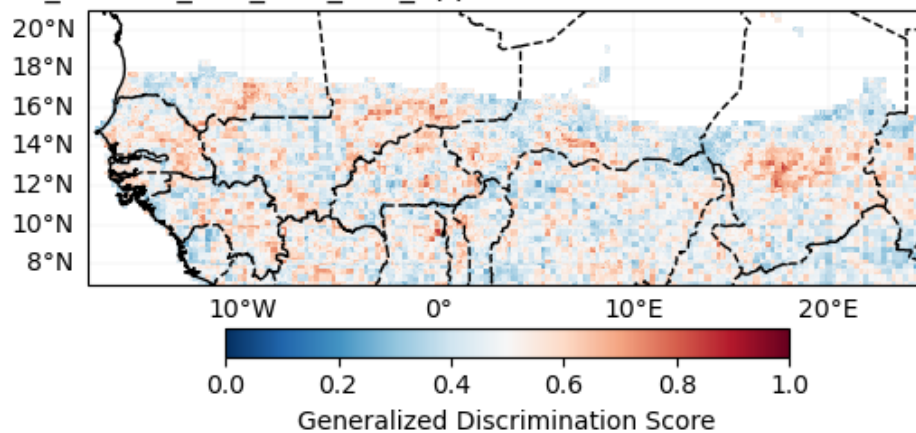
CMCC_35.VGRD_850_CCA_2nd_Approch Generalized Discrimination Score



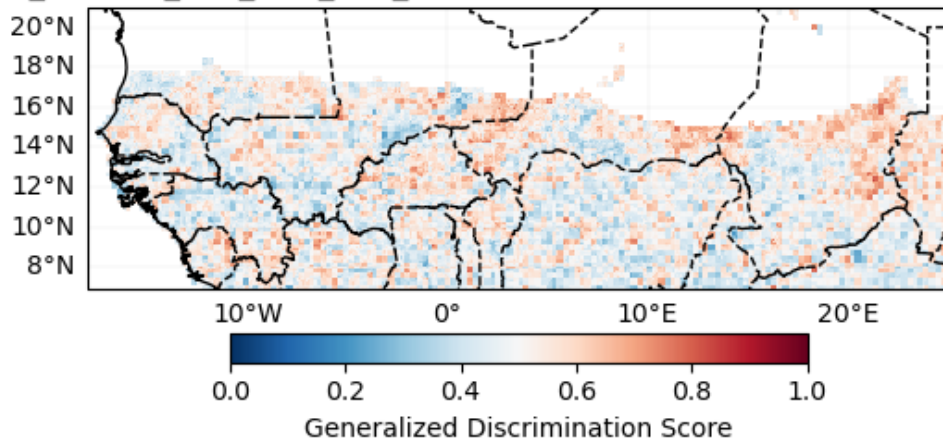
ECCC_4.VGRD_850_CCA_2nd_Approch Generalized Discrimination Score



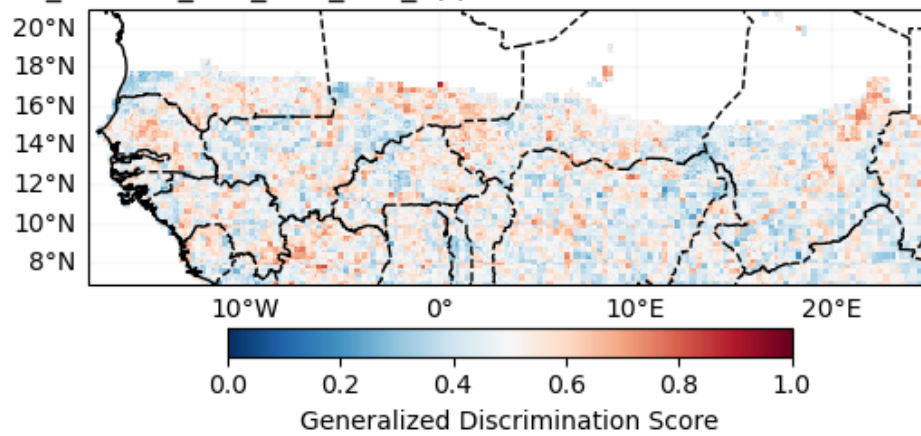
ECCC_5.VGRD_850_CCA_2nd_Approch Generalized Discrimination Score



JMA_3.VGRD_850_CCA_2nd_Approch Generalized Discrimination Score



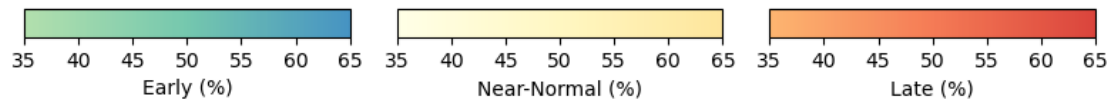
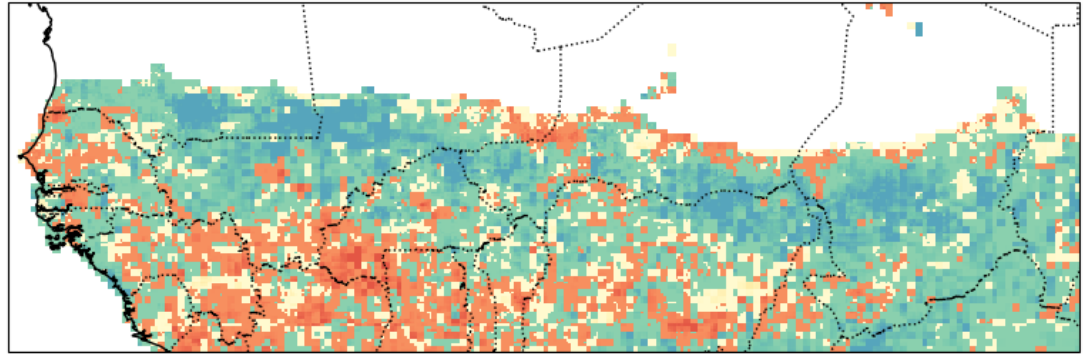
NCEP_2.VGRD_850_CCA_2nd_Approch Generalized Discrimination Score



Forecast (individual)

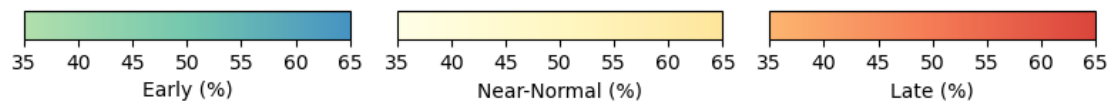
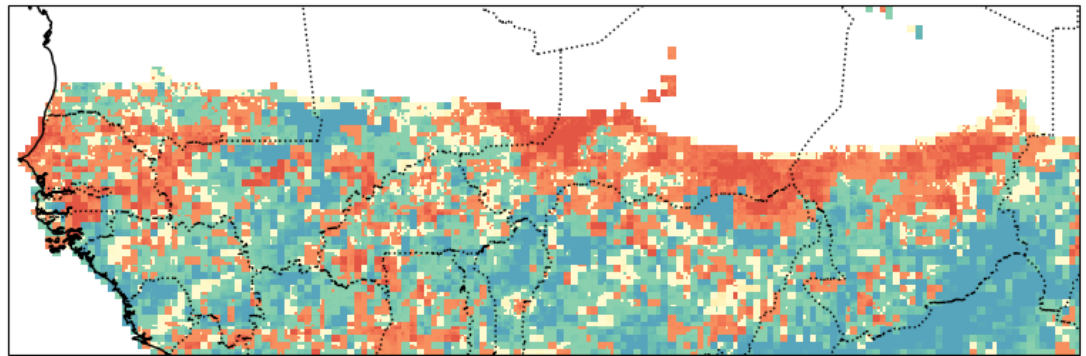
ECMWF_51

Probabilistic Forecast - CCA_2nd_Approch Onset-2025



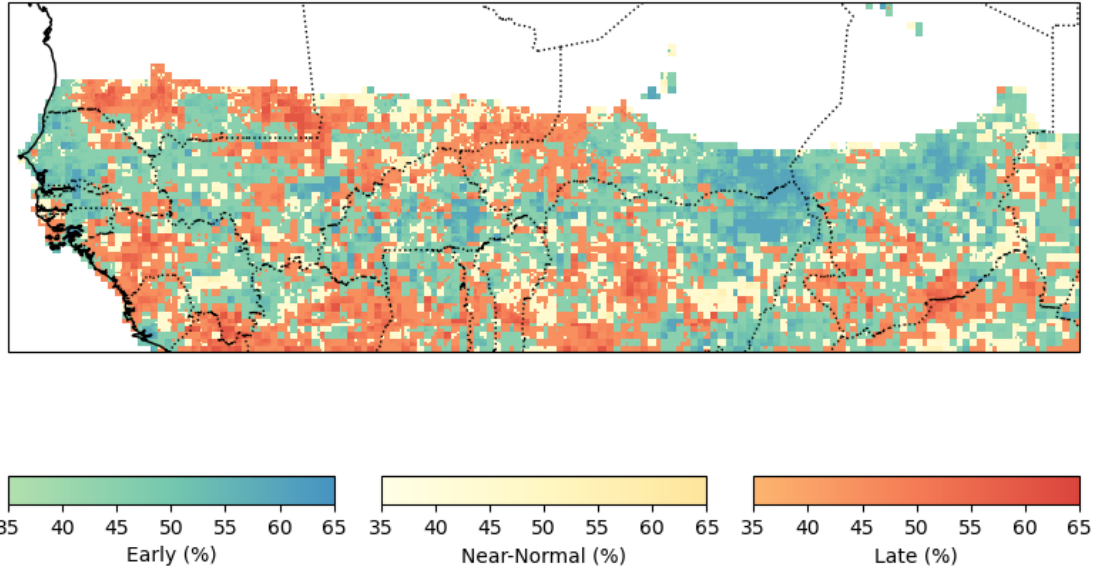
METEOFRACTANCE_8

Probabilistic Forecast - CCA_2nd_Approch Onset-2025



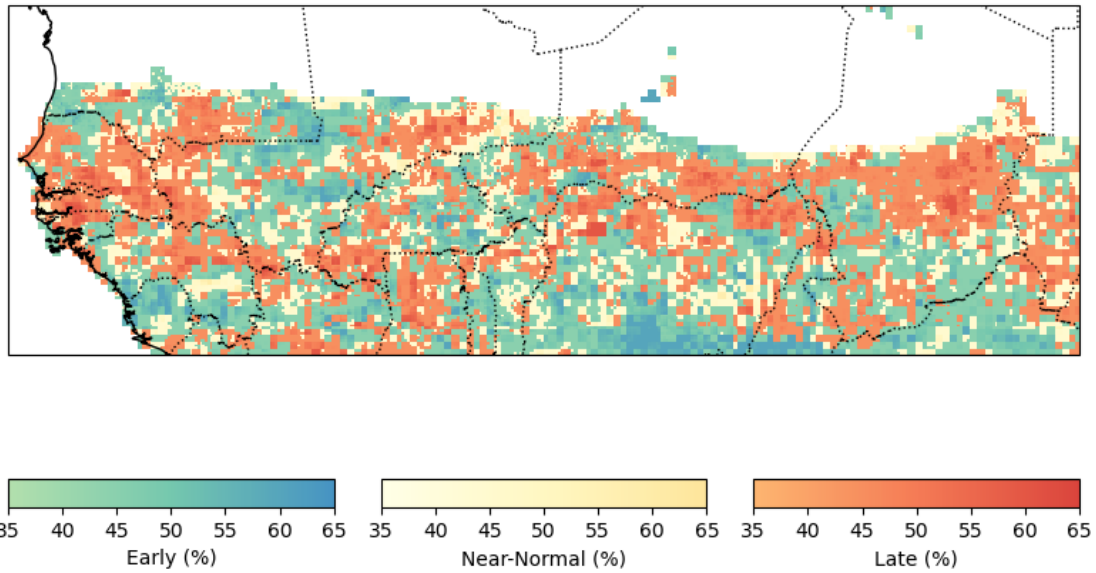
UKMO_604

Probabilistic Forecast - CCA_2nd_Approch Onset-2025



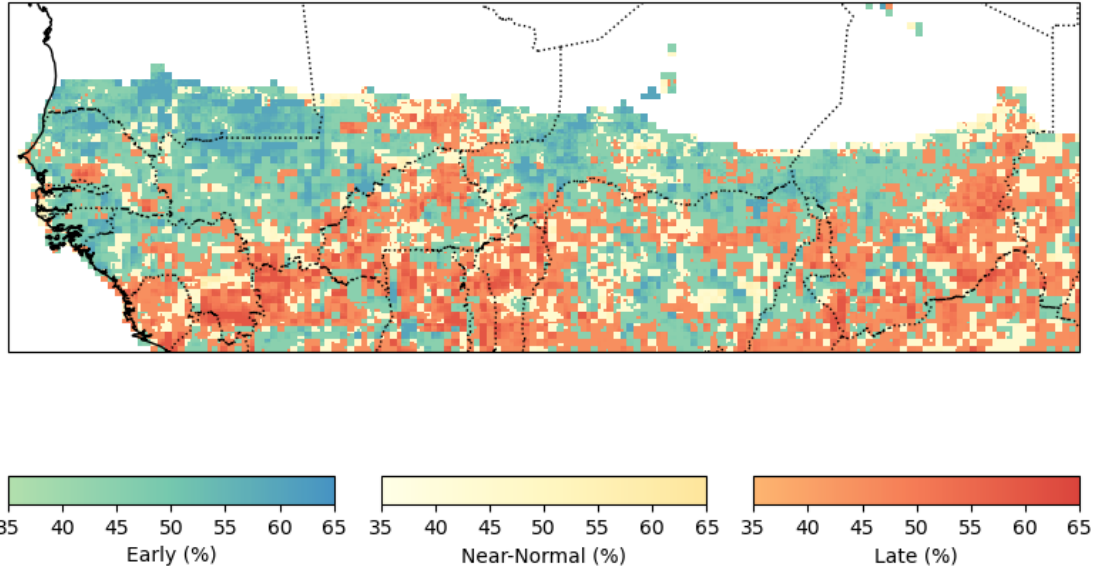
DWD_22

Probabilistic Forecast - CCA_2nd_Approch Onset-2025



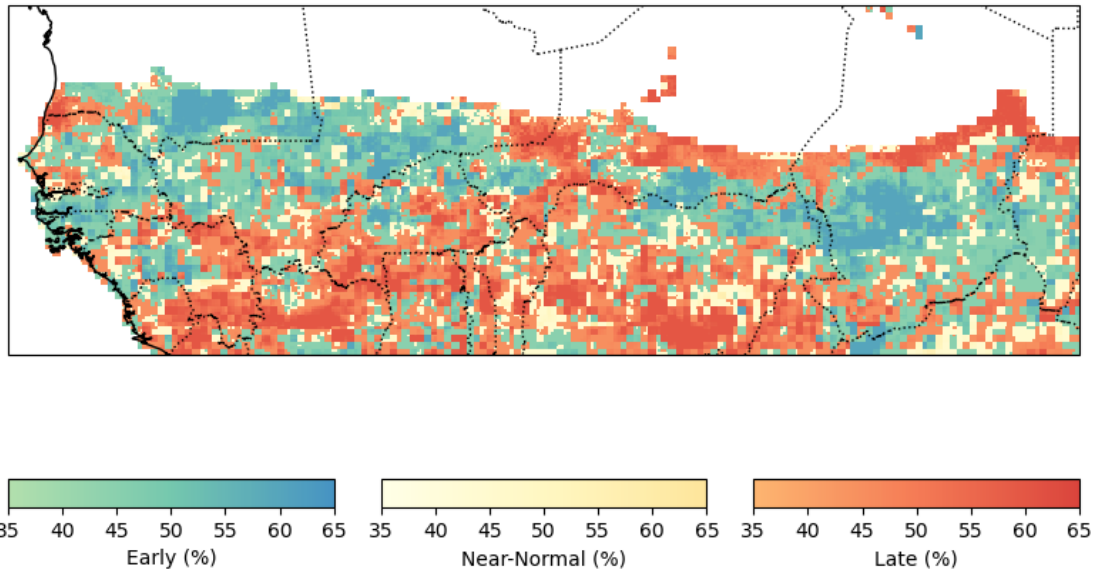
CMCC_35

Probabilistic Forecast - CCA_2nd_Approch Onset-2025



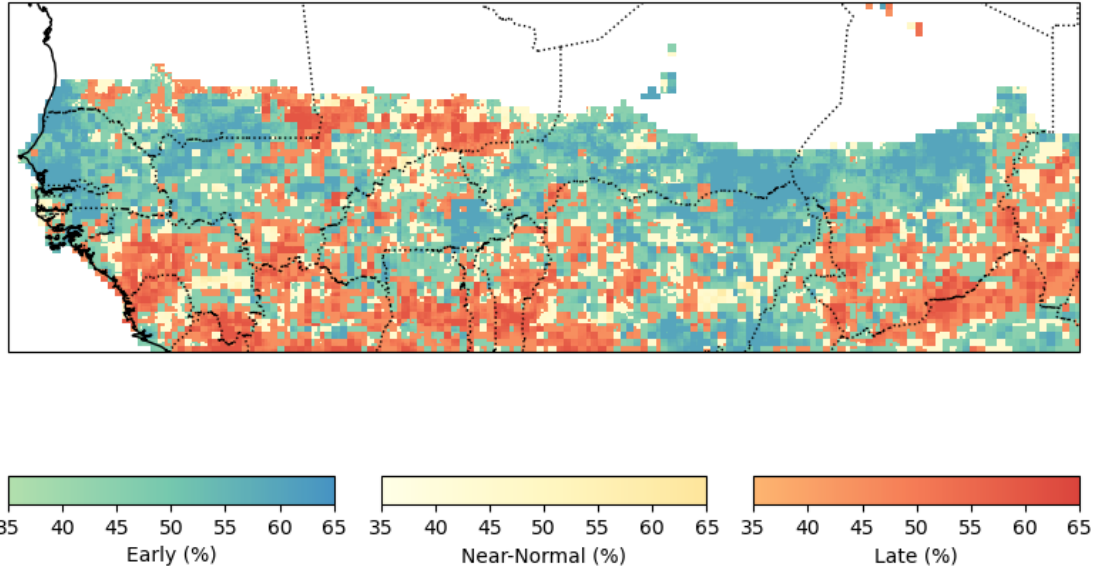
ECCC_4

Probabilistic Forecast - CCA_2nd_Approch Onset-2025



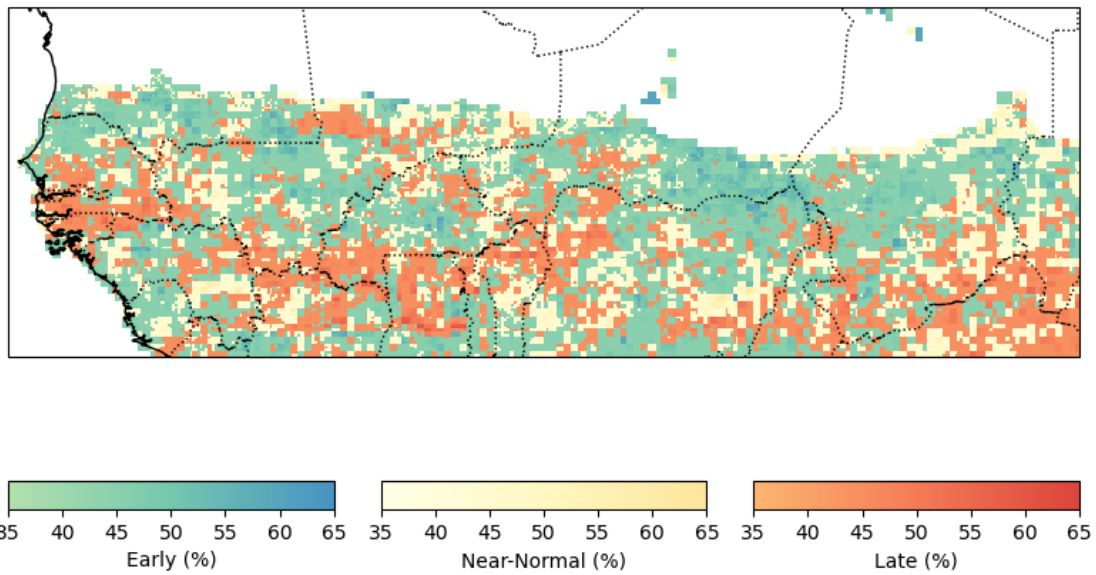
ECCC_5

Probabilistic Forecast - CCA_2nd_Approch Onset-2025



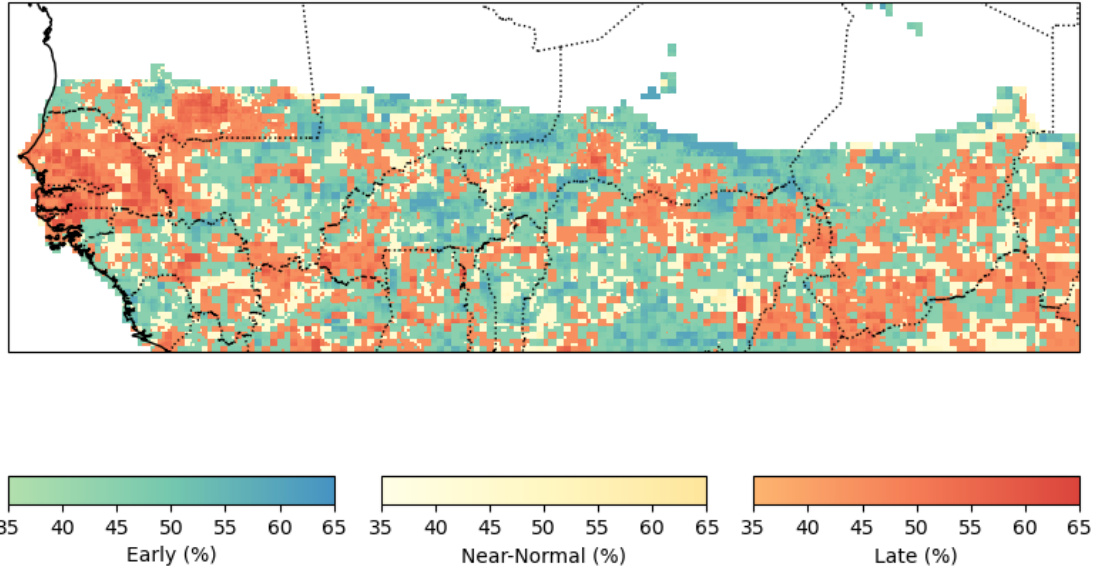
JMA_3

Probabilistic Forecast - CCA_2nd_Approch Onset-2025



NCEP_2

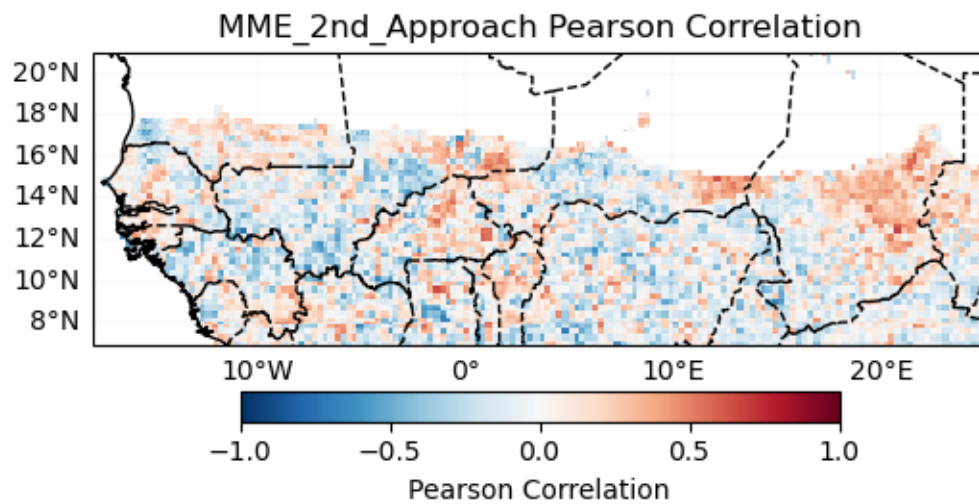
Probabilistic Forecast - CCA_2nd_Approach Onset-2025

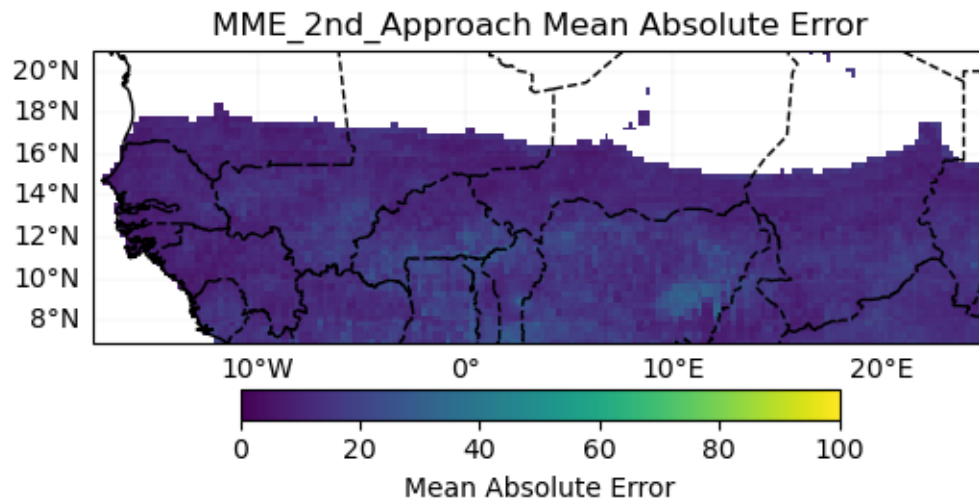


2.0.2 MME 2nd-Approach: Statistico-Dynamical

```
[ 'JMA_3.VGRD_850',
  'DWD_22.VGRD_850',
  'METEOFRACTANCE_8.VGRD_850',
  'UKMO_604.VGRD_850']
```

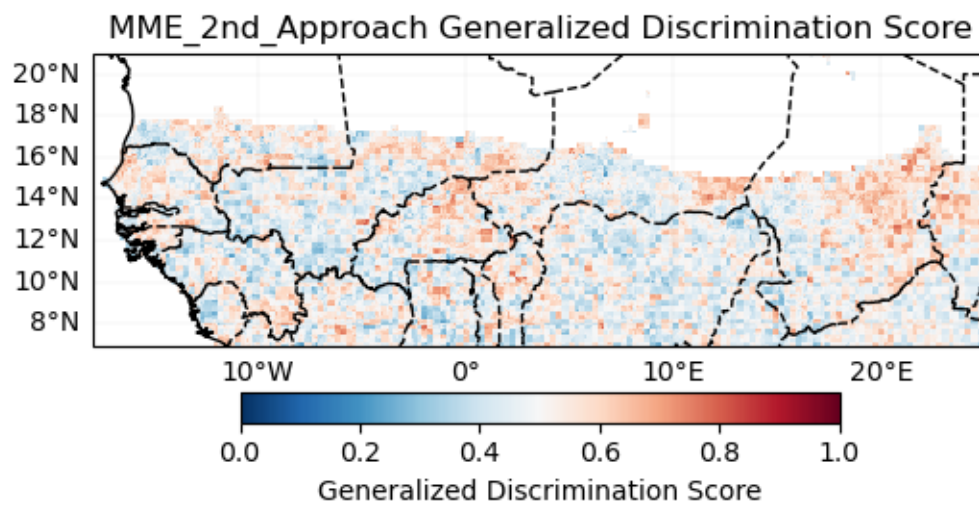
Verification Deterministic





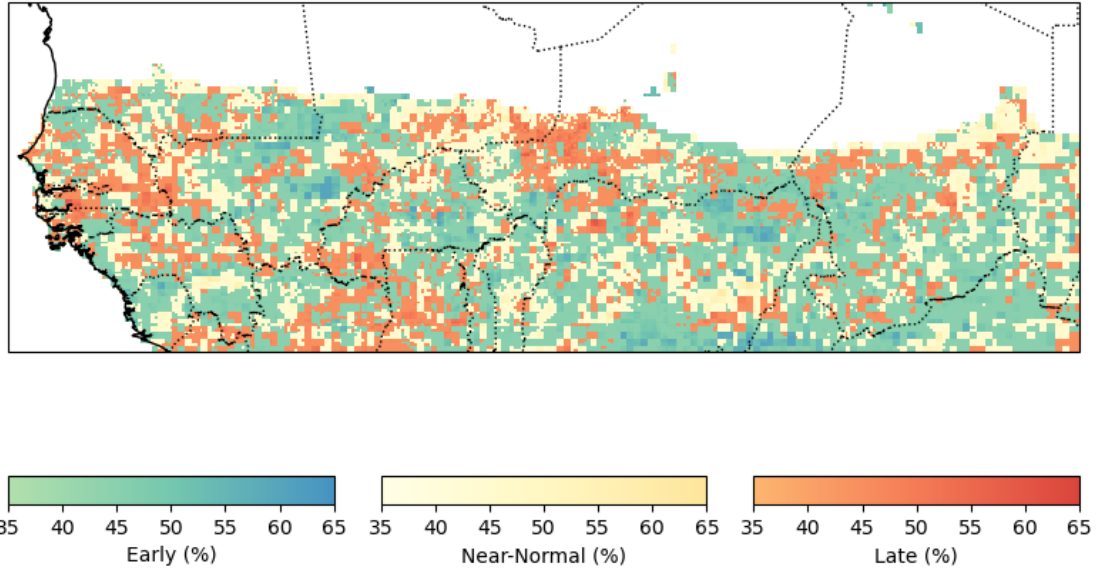
Probabilistic

Generalized Discrimination Score



Forecast

Probabilistic Forecast - MME_2nd_Approach Onset-2025



152307

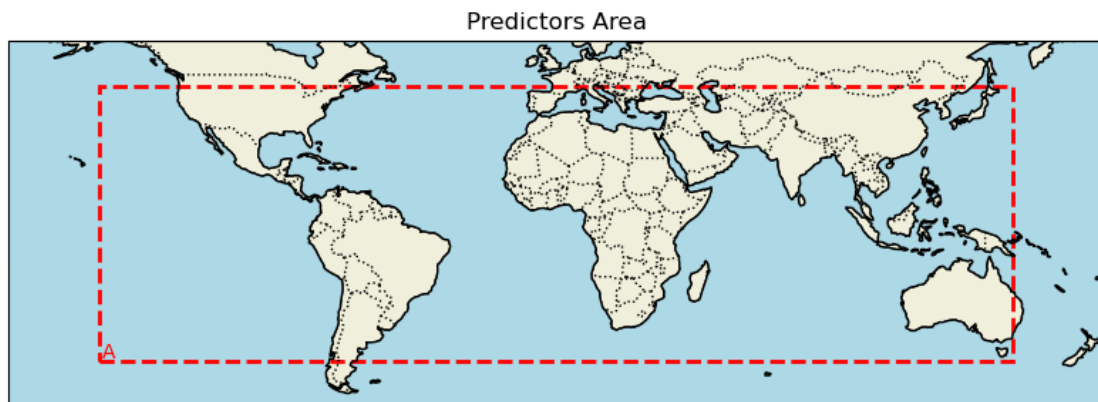
3 3rd - Approach: Observational-based seasonal forecast

Agro_PRESASS_2025/reanalysis_data/ERA5_SST_1991_2025_JanFebMar.nc already exists. Skipping.

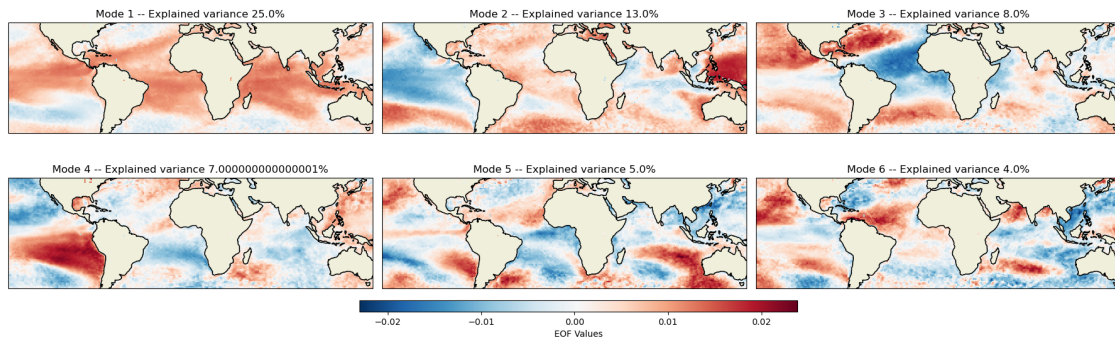
8

3.1 Initialize Principal Component Regression with ElasticNet

Zone for EOFs



EOF Modes



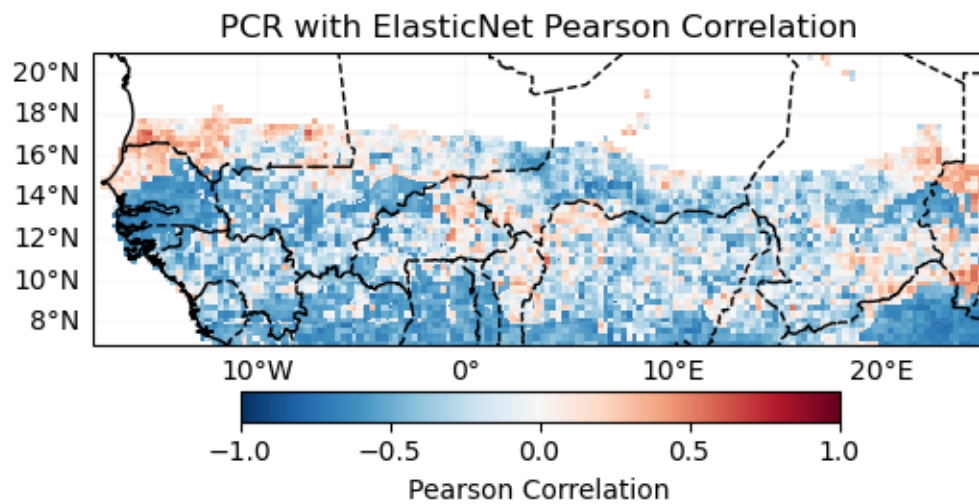
10479

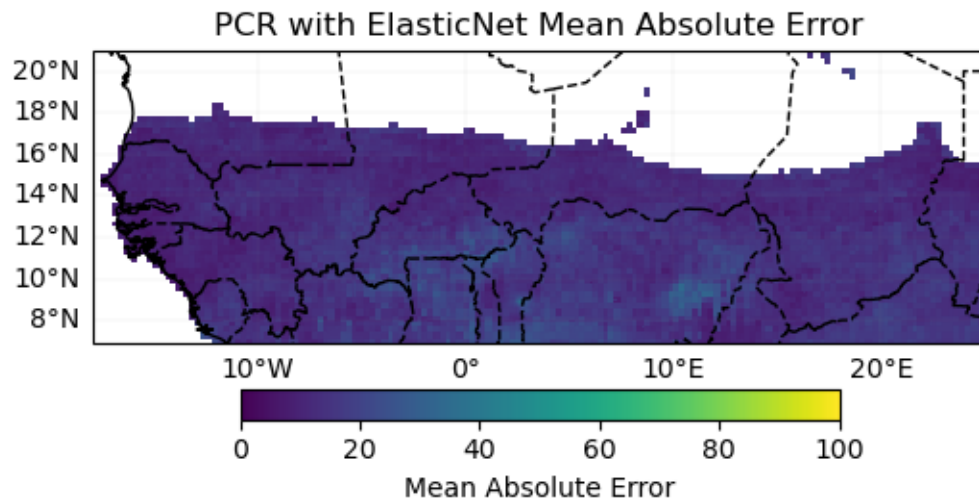
Cross-validation

Cross-validation ongoing

100% | 34/34 [07:10<00:00, 12.67s/it]

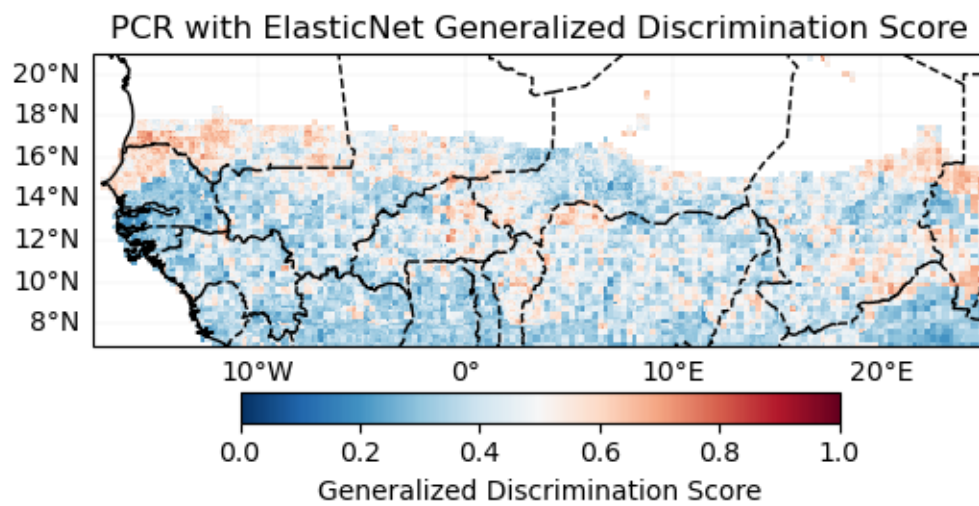
Verification Deterministic





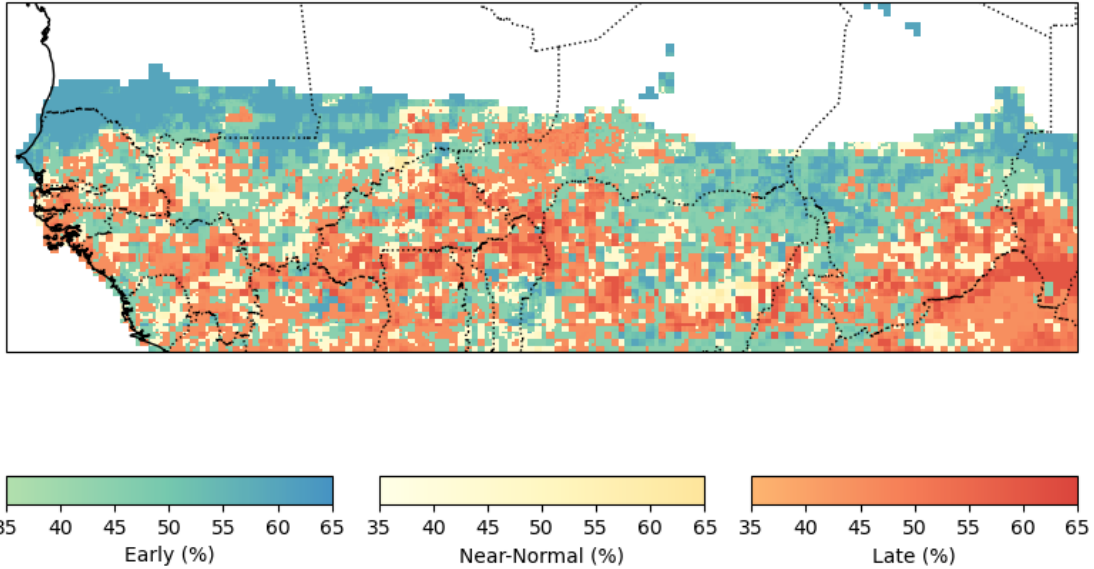
Probabilistic

Generalized Discrimination Score



Forecast

Probabilistic Forecast - PCR with ElasticNet Onset-2025



4 4th - Analog-based seasonal forecast

4.1 Initialize WAS_Analog class

Cross-validation

Agro_PRESASS_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc already exists. Skipping download.

Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization month

Agro_PRESASS_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc

File 'Agro_PRESASS_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc'

already exists. Skipping download.

All requested SST data has been processed.

Cross-validation ongoing

```
3%|                                     | 1/34 [00:00<00:06,  5.38it/s]
similar years for 1991 are [2002 1997 2023]

6%|                                     | 2/34 [00:00<00:05,  5.41it/s]
similar years for 1992 are [2020 2019 2005]

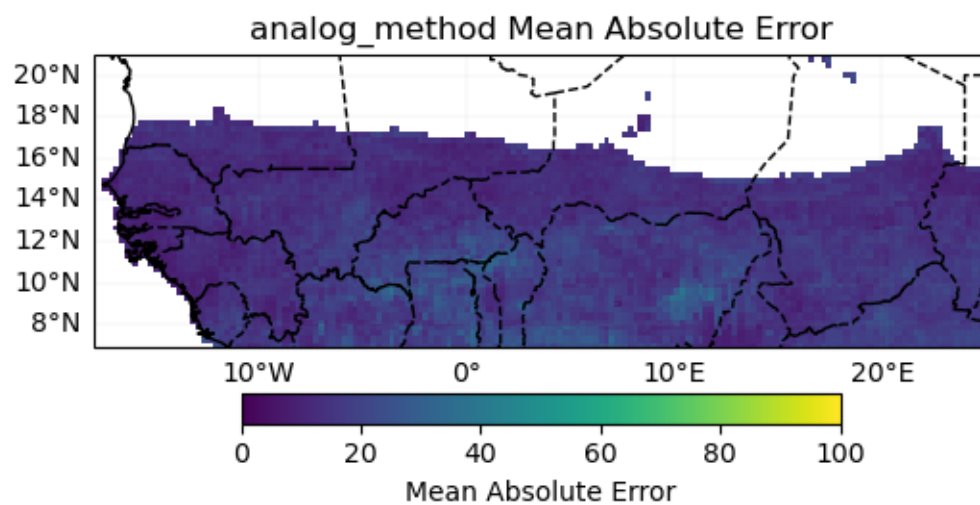
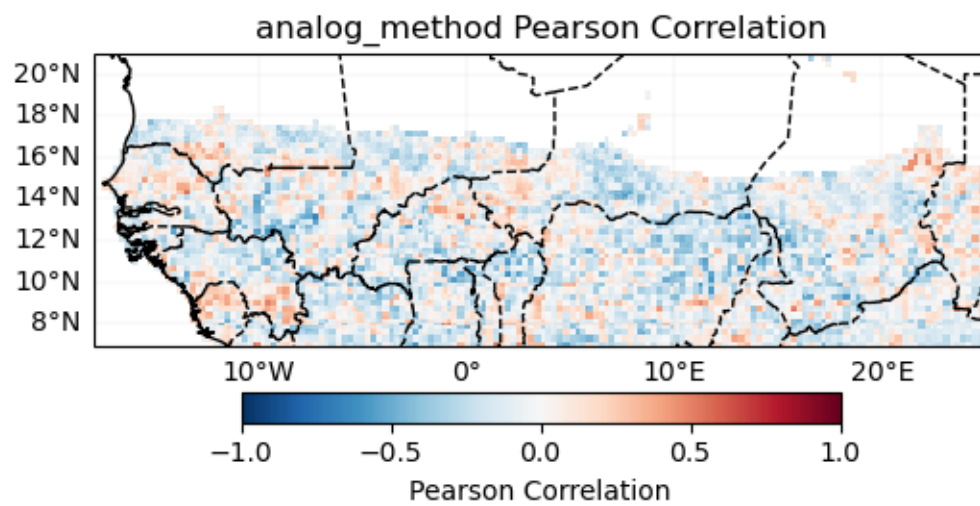
9%|                                     | 3/34 [00:00<00:05,  5.44it/s]
similar years for 1993 are [2020 1991 1996]

12%|                                    | 4/34 [00:00<00:05,  5.46it/s]
```

similar years for 1994 are [2003 1991 2009]
15%| | 5/34 [00:00<00:05, 5.47it/s]
similar years for 1995 are [2016 1998 2020]
18%| | 6/34 [00:01<00:05, 5.48it/s]
similar years for 1996 are [1993 2018 2002]
21%| | 7/34 [00:01<00:04, 5.46it/s]
similar years for 1997 are [2023 2015 2009]
24%| | 8/34 [00:01<00:04, 5.47it/s]
similar years for 1998 are [2010 2016 2024]
26%| | 9/34 [00:01<00:04, 5.47it/s]
similar years for 1999 are [2021 2011 2008]
29%| | 10/34 [00:01<00:04, 5.49it/s]
similar years for 2000 are [2012 2011 2009]
32%| | 11/34 [00:02<00:04, 5.49it/s]
similar years for 2001 are [2023 2012 2017]
35%| | 12/34 [00:02<00:04, 5.49it/s]
similar years for 2002 are [2023 1991 2018]
38%| | 13/34 [00:02<00:03, 5.49it/s]
similar years for 2003 are [2024 1998 2016]
41%| | 14/34 [00:02<00:03, 5.51it/s]
similar years for 2004 are [2007 2002 2018]
44%| | 15/34 [00:02<00:03, 5.50it/s]
similar years for 2005 are [2010 2016 2020]
47%| | 16/34 [00:02<00:03, 5.51it/s]
similar years for 2006 are [2009 2017 2011]
50%| | 17/34 [00:03<00:03, 5.52it/s]
similar years for 2007 are [2016 2010 2024]
53%| | 18/34 [00:03<00:02, 5.52it/s]
similar years for 2008 are [2021 2011 2018]
56%| | 19/34 [00:03<00:02, 5.51it/s]
similar years for 2009 are [2023 1997 2018]
59%| | 20/34 [00:03<00:02, 5.51it/s]

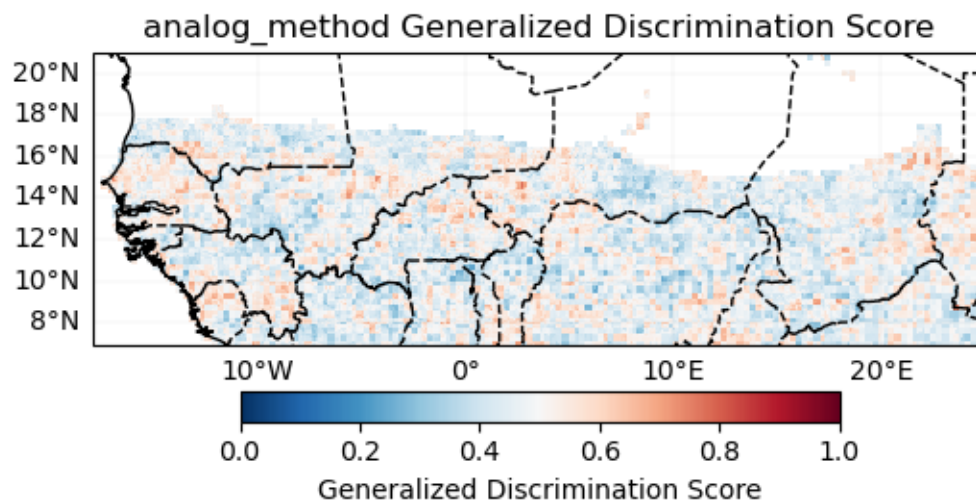
similar years for 2010 are [1998 2007 2016]
62%| | 21/34 [00:03<00:02, 5.49it/s]
similar years for 2011 are [2008 2023 2021]
65%| | 22/34 [00:04<00:02, 5.50it/s]
similar years for 2012 are [2023 2000 2011]
68%| | 23/34 [00:04<00:01, 5.52it/s]
similar years for 2013 are [2016 2020 2024]
71%| | 24/34 [00:04<00:01, 5.52it/s]
similar years for 2014 are [2023 2009 1991]
74%| | 25/34 [00:04<00:01, 5.52it/s]
similar years for 2015 are [2014 2023 1997]
76%| | 26/34 [00:04<00:01, 5.53it/s]
similar years for 2016 are [1998 1995 2007]
79%| | 27/34 [00:04<00:01, 5.55it/s]
similar years for 2017 are [2001 2023 2011]
82%| | 28/34 [00:05<00:01, 5.55it/s]
similar years for 2018 are [2021 2008 2011]
85%| | 29/34 [00:05<00:00, 5.49it/s]
similar years for 2019 are [1998 1995 2023]
88%| | 30/34 [00:05<00:00, 5.51it/s]
similar years for 2020 are [1998 2024 2016]
91%| | 31/34 [00:05<00:00, 5.51it/s]
similar years for 2021 are [2008 1999 2018]
94%| | 32/34 [00:05<00:00, 5.54it/s]
similar years for 2022 are [2018 2010 2011]
97%| | 33/34 [00:05<00:00, 5.55it/s]
similar years for 2023 are [1997 2012 2011]
100%| | 34/34 [00:06<00:00, 5.50it/s]
similar years for 2024 are [1998 2020 2003]

Verification Deterministic



Probabilistic

Generalized Discrimination Score



Forecast

Agro_PRESASS_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc already exists. Skipping download.

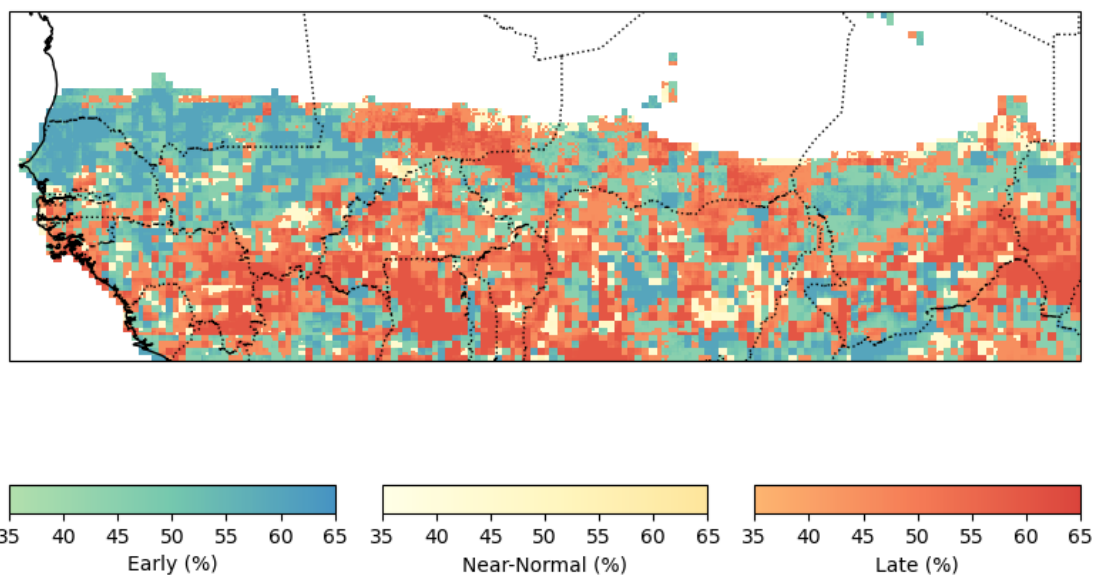
Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization month

Agro_PRESASS_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc

File 'Agro_PRESASS_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc' already exists. Skipping download.

All requested SST data has been processed.

Probabilistic Forecast - analog_method Onset-2025



Agro_PRESASS_2025/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc already exists. Skipping download.

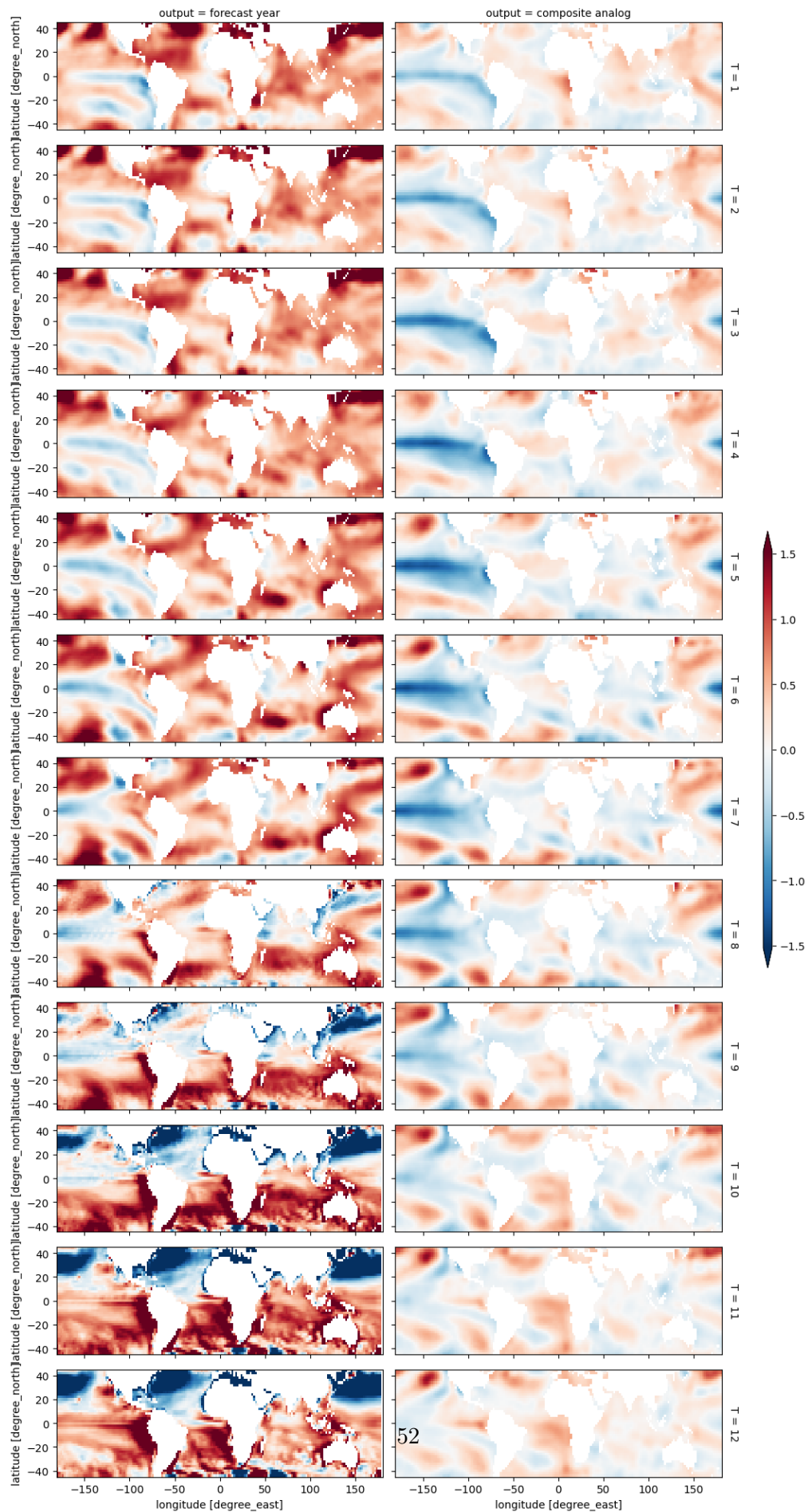
Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization month

Agro_PRESASS_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc

File 'Agro_PRESASS_2025/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc' already exists. Skipping download.

All requested SST data has been processed.

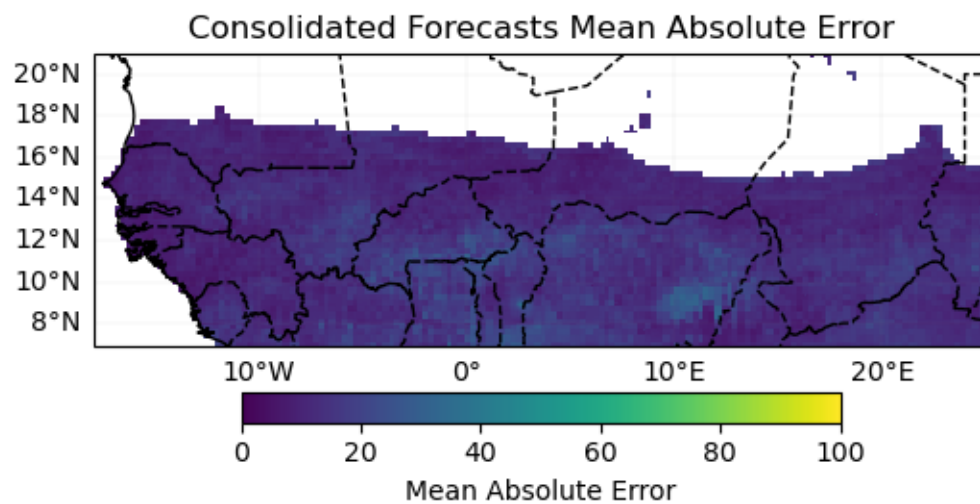
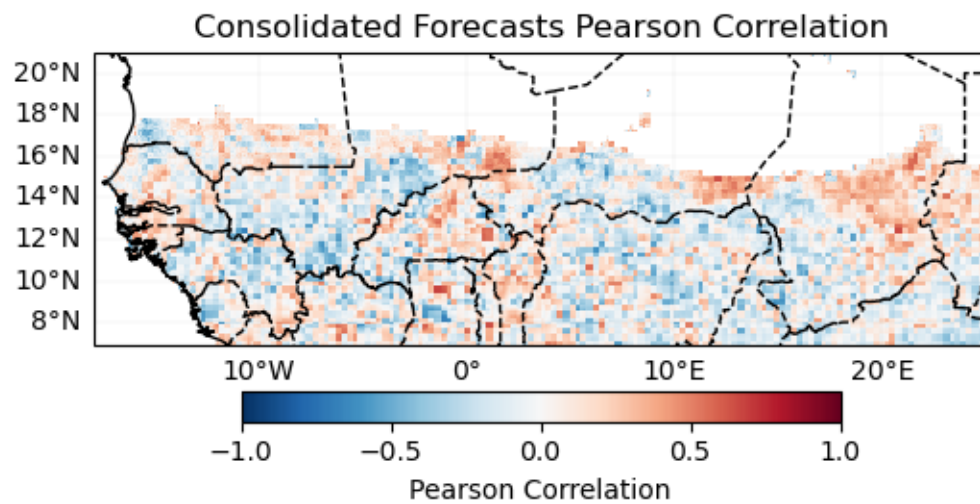
array([2021, 2009, 2008])



5 Consolidated Forecasts

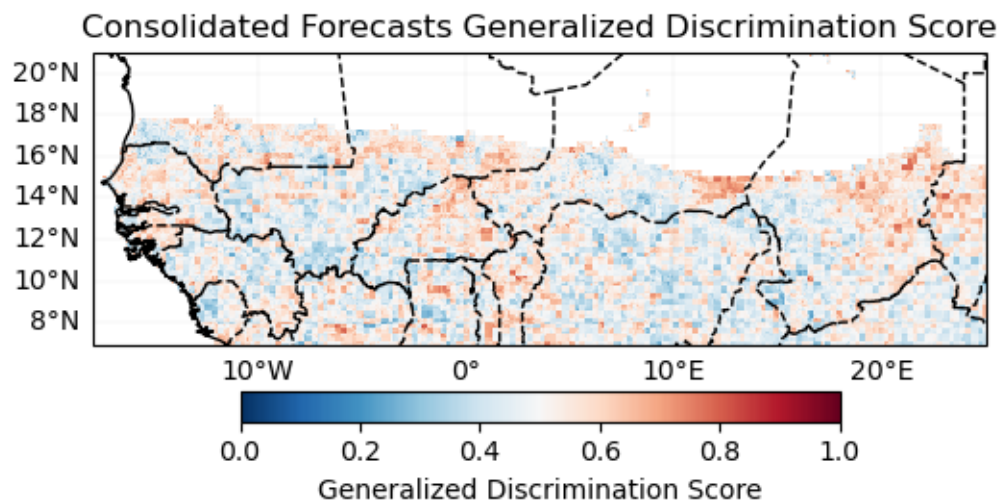
Cross-validation

Verification Deterministic

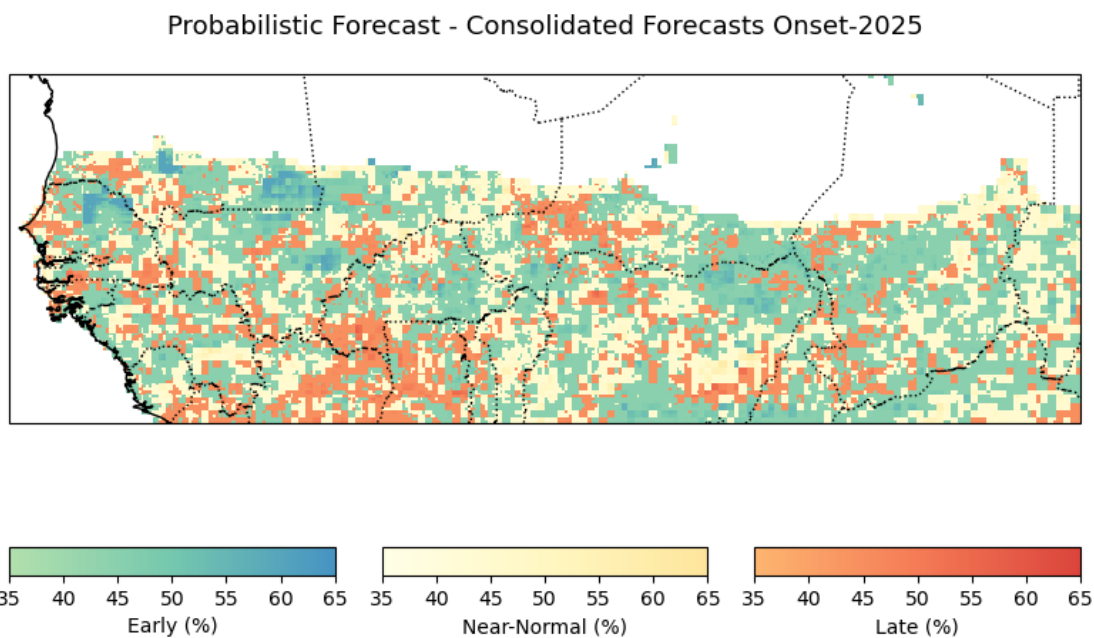


Probabilistic

Generalized Discrimination Score



Forecast



24785

Save Forecast for future evaluation

5.0.1 --- - END ---

```
[NbConvertApp] Converting notebook ./01_MME_Onset.ipynb to pdf
[NbConvertApp] Support files will be in 01_MME_Onset_files/
[NbConvertApp] Making directory ./01_MME_Onset_files
[NbConvertApp] Writing 65769 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 12518780 bytes to 01_MME_Onset.pdf
```